

Abstract¹

Latin America is aging before it has grown rich. We ask what that costs, who pays, and what the standard pension reforms buy, in a stochastic overlapping-generations economy estimated for Mexico. Health is endogenous capital raising survival, productivity, and the value of consumption; agents differ by sex and education, each type carrying its own measured earnings and survival schedules. Factor prices and the pay-as-you-go contribution rate clear in equilibrium, so demography reaches households as both a price and a tax. The remaining parameters are estimated on the age profile and income gradient of household medical spending, the skill premium, and mid-life survival by education. Comparing stationary equilibria, the 2050 demographics more than double the contribution rate, from 5.66 to 12.26 per cent, yet leave a newborn almost exactly as well off: slower population growth deepens capital, and the higher wage and pension nearly offset the heavier wedge. The cost falls instead on those already holding assets, and rises steeply with age. Against inaction, cutting benefits by 54 per cent is worth 8.3 per cent of lifetime consumption to a newborn, and working to seventy 3.1 per cent; every type ranks them alike, and the ranking reverses with age. The margins differ far less in the welfare they deliver than in whose old age pays.

JEL Classification: E62, H55, I13, J11, J26

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1 Introduction

Latin America is aging before it has grown rich. The demographic transition that took Europe some seventy years will take the region about twenty-five (Fietz et al., 2025), and it arrives in economies with incomplete formal labor markets and little fiscal room to absorb it (Pessino and Ter-Minassian, 2021). México makes the problem concrete. Its old-age dependency ratio is projected to more than double between 2020 and 2050, from 0.141 to 0.307 (CONAPO, 2023), and a pay-as-you-go pension system has only a few margins on which to absorb a shift of that size. What does the adjustment cost? Who bears it? And what do the classical reforms, cutting benefits or working longer, buy relative to doing nothing? This paper answers those three questions for México.

We build a stochastic overlapping-generations economy in which health is an endogenous capital stock that raises survival, labor productivity, and the value households place on consumption. Agents come in four types, two education groups for each sex, and every type carries its own measured efficiency and survival schedules, so the differences that turn out to decide who pays enter the model as data rather than as assumptions. Factor prices clear, and so does the pension system's contribution rate, which means demographic pressure reaches households as an equilibrium outcome rather than as an imposed tax. We estimate the model on Mexican micro-data covering the age profile and income gradient of household medical spending, the skill premium, and mid-life survival by education, drawn from the ENIGH household surveys and the ENASEM aging panel. Every experiment then runs at the estimated baseline's own general-equilibrium solve, so each counterfactual perturbs an economy the data have disciplined.

Aging under unchanged policy turns out to redistribute rather than to impoverish. Imposing the 2050 demographics and leaving policy alone, the equilibrium contribution rate more than doubles, from 5.66 to 12.26 per cent, and yet a newborn is left almost exactly as well off as before. Capital deepening is the reason. Slower population growth tilts the age distribution toward older and asset-rich households, so the annual return falls from 8.58 to 6.82 per cent while the wage rises 12.4 per cent and the wage-indexed pension almost 20 per cent. For someone who arrives holding nothing, those gains very nearly pay the doubled wedge. The cost lands instead on households that already hold assets, whose remaining lifetimes are repriced at the lower return, and it climbs steeply with age: one to two per cent of remaining consumption at ages 25–29, six to eight per cent at 40–44, and twelve to twenty-one per cent at 65–69.

The identity that closes a pay-as-you-go system leaves three margins, the contribution rate, the replacement rate, and the boundary between contributors and beneficiaries, and the fiscal arithmetic on each is available before any equilibrium is solved. Inaction takes the contribution rate to 12.26 per cent. Holding it at its 2020 value instead requires cutting the replacement rate from 0.40 to 0.185, a 54 per cent benefit cut. Raising the retirement age from 65 to 70 lands it at 8.78 per cent. What general

equilibrium adds is the welfare content the identity cannot supply. Measured against inaction, the benefit cut is worth 8.3 per cent of lifetime consumption to a newborn and the later retirement 3.1 per cent, a ranking all four types agree on and one that reverses with age. Where the reforms differ most is in whose old age pays: inaction taxes wealth-holders through the return, the benefit cut taxes pension claims, and the later retirement taxes the extended work band. One qualification governs all three. These are stationary economies, so the numbers price destinations rather than the journeys between them, and the transition generations of any reform are not modeled.

Which type pays is decided by heterogeneity the model measures rather than assumes. Low-education men are the only meaningful newborn losers under inaction, because their education gap in survival opens some fifteen years earlier in the staged schedules than women's: they pay the doubled contribution across a full working life and collect the raised pension for the fewest years. Low-education women bear the higher retirement age, because their estimated late-life efficiency is the lowest of the four types, so the extended work band makes them work at wages that barely compensate the effort. We flag the fragility of the first pattern where we report it, but neither statement is available at all in a model without sex- and education-specific survival and efficiency schedules.

The sex and education structure behind those statements is measured rather than fitted. The efficiency and survival schedules enter from the first step, so the model carries the observed gaps into its equilibrium instead of being asked to match them, and the age profile of the gender earnings gap it produces is a property of that staging rather than a test of the model. Where the model is genuinely tested it holds up: its willingness to pay for survival, which no moment targets, comes in about a sixth above an external benefit-transfer benchmark. At the estimates, however, the endogenous investment channel generates little of the observed education-survival gradient, a limit we measure and discuss in Section 5.

Three contributions follow. The first and principal one is economic: the general-equilibrium incidence of demographic aging and of the three pension margins, priced at an estimated baseline for an emerging economy. The second is the apparatus, a calibrated overlapping-generations economy for México in which health, longevity, earnings, and the pension system's finances are jointly determined. The third is the heterogeneity that apparatus carries. Quantitative models of pension reform ordinarily resolve incidence by age and by wealth; this one resolves it by sex and by education as well, on schedules estimated separately for each of the four groups, and that is what decides who bears each reform. In doing so we extend the quantitative overlapping-generations literature on aging and pension reform to an emerging economy, and extend it along a dimension that literature has largely left open. Section 2 places the paper against work on aging and pension reform, on health as capital, on health financing in Latin America, and on gender heterogeneity in lifecycle fiscal models.

The remainder of the paper proceeds as follows. Section 2 reviews the related liter-

ature. Section 3 presents the model and defines the stationary equilibrium. Section 4 reports the two-step Mexican calibration and the estimation. Section 5 presents the evidence on Mexican medical spending and what the estimated health block delivers. Section 6 develops the pension experiments at the estimated baseline, and Section 7 concludes.

2 Literature Review

This paper draws on four literatures. The first is the quantitative overlapping-generations literature on demographic aging and pension reform, which supplies the questions the paper answers and the benchmarks against which its answers are read. The second is the literature that models health as a capital stock built by forward-looking investment, which supplies the apparatus. The third fixes the institutional environment in which health and pensions are financed in Latin America and bounds what the model can be asked. The fourth locates the heterogeneity by sex and education through which every incidence result in the paper is priced. They are not treated at equal depth: the review is weighted toward the first, second, and fourth, where the paper's contributions lie.

2.1 Aging, Pension Reform, and Incidence in Quantitative Overlapping-Generations Models

Since [Auerbach and Kotlikoff \(1987\)](#), the workhorse for analyzing the intergenerational fiscal consequences of demographic change has been the multi-period overlapping-generations model with a government sector, and three threads of that literature organize this paper's principal results.

The first is the transmission of aging to welfare through factor prices. [Krueger and Ludwig \(2007\)](#) establish the central result: population aging raises the capital-labor ratio, so wages rise and returns fall, and the welfare consequences split by what a household holds—agents whose lifetime resources are mostly labor income gain, while cohorts already holding assets lose as their wealth is repriced at the lower return. This paper's first finding is that mechanism, obtained for México. Where [Krueger and Ludwig \(2007\)](#) show that the adjustment of the contribution rate can offset the direct effect of aging, we price that offset at an estimated emerging-economy baseline and resolve its incidence by age, sex, and education: because our pay-as-you-go contribution rate clears endogenously, aging arrives at the household simultaneously as a factor-price change and as a doubled tax wedge, and the newborn's near-indifference is the net of the two. [Carvalho et al. \(2016\)](#) separate the channels through which aging moves the equilibrium return—longevity and the dependency ratio—a decomposition directly relevant here, since our demographic experiment moves fertility only and holds sur-

vival at its estimated schedules. [Auclert et al. \(2021\)](#) quantify these forces at the frontier and across countries, and their finding that the sign and size of the aging effect depend heavily on country characteristics is the reason an emerging-economy calibration is not a mechanical application of the high-income results.

The second thread is the reform menu itself. [Kitao \(2014\)](#) enumerates the options for restoring pay-as-you-go sustainability under aging—raising the payroll tax, cutting the replacement rate, raising the retirement age, and means-testing benefits—and prices them in a general-equilibrium life-cycle model. Our menu is Kitao’s minus means-testing, which the flat benefit of Section 3 cannot represent, and the omission is a property of the benefit rule rather than a judgment about the instrument. That the choice among pension architectures is fundamentally distributional, and can be posed as a comparison across stationary economies, is established by [Huggett and Ventura \(1999\)](#); their exercise is the precedent for our own restriction to steady states. Two papers supply the counterweight that disciplines how our welfare rankings may be read. [Conesa and Krueger \(1999\)](#) and [Nishiyama and Smetters \(2007\)](#) establish that an unfunded pillar provides insurance against uninsurable idiosyncratic wage risk: [Nishiyama and Smetters \(2007\)](#) show that the efficiency gains from shrinking it reverse once that insurance is priced, and [Conesa and Krueger \(1999\)](#) that it is what sustains political support for the status quo.

The third thread is differential mortality. [Sánchez-Romero et al. \(2020\)](#) show that longevity gaps by socioeconomic status transfer resources toward the long-lived and undermine the actuarial fairness of pension systems; it is the closest existing work to this paper’s finding that the education gradient in survival decides which type loses under inaction. [Bagchi \(2019\)](#) supplies the caution, finding that optimal progressivity is surprisingly insensitive to wealth-dependent mortality—a result worth holding against ours, which reports an incidence pattern rather than an optimal policy.

Two further anchors bound the exercise. Every welfare ranking below rests on the condition of [Aaron \(1966\)](#): where the market return exceeds the growth rate of the contribution base, mandatory participation in an unfunded pillar transfers resources away from the participant. Whether that condition holds is not settled by assumption, and [Abel et al. \(1989\)](#) show that the comparison is delicate once uncertainty is admitted; in the deterministic aggregate environment of this model the gap is direct, and the annual return ranges from 5.89 to 8.58 percent against base growth of at most 3.51 percent, so the condition holds with room to spare in every economy on the menu. And the factor-price channel that carries our first finding is a closed-economy object. [Attanasio et al. \(2007\)](#) show that capital mobility between asynchronously aging regions substantially mitigates the factor-price consequences of aging, and that the South’s exposure differs from the North’s; we treat the closed-economy assumption as this paper’s most fragile and flag it where the results depend on it. For the regional fiscal scale of the problem, [Pessino and Ter-Minassian \(2021\)](#) project pension and health spending pressure across Latin America over a fifty-year horizon.

This paper’s contribution to this literature is to price all three feasible margins side by side, in one estimated general equilibrium for an emerging economy, with incidence resolved by age, sex, and education through measured schedules rather than assumed heterogeneity. One bound on that contribution should be stated here rather than discovered later. Our welfare comparisons price the return gap of the Aaron condition between stationary economies, but they do not price the journey between them: the newborn gain we report from cutting benefits accrues to a newborn of the reformed steady state who never contributed at the old rate, and on the evidence of [Nishiyama and Smetters \(2007\)](#), who compensate transitional losses explicitly, it should be read as an upper bound on what that reform is worth.

2.2 Health, Longevity, and the Life Cycle

The apparatus this paper builds descends from [Grossman \(1972\)](#), and the inheritance should be stated as an assumption rather than absorbed as a convention: health is a depreciable capital stock, medical spending is forward-looking investment in that stock, and the stock so built governs survival and—in the richer versions—labor productivity. Four functions have been asked of that block, and the literature is best read by which one a given paper assigns it.

The first is to generate aging and mortality endogenously. [Dalgaard and Strulik \(2014\)](#) derive the pattern of aging and death from the interaction of health investment with biological depreciation, yielding a structural reading of the Preston curve; [Kuhn and Feichtinger \(2015\)](#) analyze health and retirement choices jointly when health capital depreciates with age and can be partially restored; and [Fioroni \(2010\)](#) shows that the interaction of precautionary saving with health investment produces consumption–saving profiles unlike those of models with exogenous health. The second function is to price life itself. [Hall and Jones \(2007\)](#) show that when health enters preferences so that the marginal value of additional life rises with consumption, the aggregate income elasticity of health spending exceeds one and health’s expenditure share grows with development; [Murphy and Topel \(2006\)](#) develop the same valuation structure and its empirical magnitudes. This paper’s multiplicative health amenity, and the income gradient in willingness to pay for health it generates, is that mechanism embedded in a general-equilibrium life cycle.

The third function is to close a macroeconomic loop. [Jung and Tran \(2016\)](#) and [Jung et al. \(2017\)](#) build stochastic general-equilibrium overlapping-generations economies with endogenous health capital and health insurance; [Jung et al. \(2017\)](#) use that apparatus for aging experiments—the closest existing statement of the apparatus combination this paper claims, executed for the United States. [Zhao \(2014\)](#) closes the loop in the other direction, showing that social security itself alters health investment and longevity, while [Franković and Wrzaczek \(2020\)](#) trace medical progress through capital accumulation and factor prices, [Yew and Zhang \(2018\)](#) introduce longevity ex-

ternalities, [Lim \(2020\)](#) shows that public health insurance reshapes aggregate saving in Korea, and [Carrasco \(2026\)](#) operates the same block as a macroeconomic transmission channel in which investment in health lengthens lives, strengthens life-cycle saving, and moves the equilibrium interest rate. The fourth function is to carry expenditure risk. [De Nardi et al. \(2010\)](#) show that uncertain medical expenses are a primary driver of late-life saving among older American households; [Hosseini et al. \(2022\)](#) estimate the health-deficit process on United States panel data with health observed from early adulthood; and [Gertler and Gruber \(2002\)](#) establish for a developing economy the fact that motivates the whole design—major illness is a large shock that households insure poorly.

The evidentiary base on which the block has been estimated is narrower than its use. These models are calibrated or estimated overwhelmingly on high-income data, in economies where coverage is broad and out-of-pocket payments are a modest share of health spending, so the data do not force a distinction between spending that builds the stock and spending that repairs it after a shock. Where that distinction has been confronted, it has been confronted in partial equilibrium and on United States data: [Ozkan \(2025\)](#) builds a life-cycle health-capital model that separates preventive from curative spending and documents how their income gradients differ across the life cycle, and [Galama and van Kippersluis \(2013\)](#) argue that the standard criticisms of the Grossman framework rest on a degenerate special case that reformulations resolve. What has not been attempted is that confrontation inside a general equilibrium, or on data from an economy in which out-of-pocket payment is the dominant mode of health financing.

The paper’s position here rests on three qualifiers doing visible work, and the near-misses are cited so that a reader can check each. [Jung and Tran \(2016\)](#) and [Jung et al. \(2017\)](#) deliver the general equilibrium and the endogenous health stock, but for a high-income economy with broad insurance; [Zhao \(2014\)](#) delivers the health–fiscal loop, likewise for the United States; [Hosseini et al. \(2022\)](#) deliver estimation on micro-data, in partial equilibrium. What this paper adds, for the health block specifically, is the conjunction: a block carrying survival, labor productivity, and the fiscal loop at once, estimated rather than calibrated, in an economy where health spending is largely out of pocket. We claim the combination and not any of its parts—and Section 5 reports how much of the education–survival gradient the investment channel actually generates once the block is estimated.

For the Latin American context, [Cortés et al. \(2024\)](#) develop a partial-equilibrium overlapping-generations model for México in which agents differ by skill and face stochastic health transitions, health-dependent productivity, and age-varying out-of-pocket expenditure. Their results—consumption gaps of 40–70 percent between good and poor health groups, and a 20-year gap in healthy longevity by skill level—establish the magnitudes at stake in the Mexican setting; the present paper is disciplined instead against its own ENIGH and ENASEM moments. The extensions are general equi-

librium with endogenous wages, an explicit pension system whose contribution rate clears endogenously, and heterogeneity by sex as well as skill.

2.3 Health Financing, Social Insurance, and the Tax Wedge

The economics of health insurance in developing economies builds on the insight that health shocks are both large and poorly insurable through private markets, generating welfare losses that pooling arrangements can substantially mitigate (Arrow, 1963). In Latin America the design question is not whether to pool health risk but through which institutional channel. Systems in the Bismarckian tradition link coverage to formal employment; systems in the Beveridgean tradition link it to residency or citizenship; and, as Levy (2008) emphasizes, most countries in the region operate hybrids of the two, often with fragmentation, duplication, and unequal benefit packages.

Summers (1989) supplies the valuation logic that organizes the modern literature: mandated contributions are priced into wages and generate no net distortion when workers value the benefits they purchase, and they function as a tax on formal labor precisely when workers do not—because benefits are not credible, or because their arrival is too distant. Fietz et al. (2025) formalize this logic as the Formalization Tax Rate, the net fiscal cost of formal employment inclusive of forgone non-contributory benefits, and show that it is highest for low-income workers, the group with the weakest attachment to the formal sector. Within this literature, Baeza and Packard (2006) document the asymmetry most relevant to this paper’s design: health coverage delivers immediate, tangible protection against catastrophic expenditure and is the benefit workers learn most about and value most highly, while pension benefits arrive far in the future and are discounted for institutional credibility.

What this subsection establishes for the paper is a boundary rather than a contribution. The model takes a deliberate position in this landscape: its pension benefit is flat and indexed to contemporaneous average earnings, so there is no benefit–contribution link at the individual level—a Beveridgean benefit inside a Bismarckian financing shell. The simplification buys tractability in the fiscal block, and it bounds what the paper can say about formalization incentives, which respond to exactly the individual-level link the model omits. One feature is worth noting against this literature nonetheless: the wedge it studies is imposed exogenously in most applications and is an equilibrium object here, since the contribution rate clears the pension system rather than being set.

2.4 Gender and Education Heterogeneity in Lifecycle Fiscal Models

Gender-differentiated incidence in social security is an active subject in estimated life-cycle models. Borella et al. (2023) estimate a life-cycle model of couples and singles with survival risk, medical expenses, and the marriage-related provisions of United States Social Security, and show that those provisions carry large and unequal consequences

for women’s labor supply and welfare across cohorts. [Fehr and Kindermann \(2018\)](#) provide the reference framework on the household side, a two-person household whose female member’s labor supply responds to childcare, parental leave, and joint-versus-individual taxation—but in an environment without health as a state variable, without endogenous survival, and without a fiscal loop.

The natural extension of that work to this paper’s environment has not been attempted, and the reason is computational rather than conceptual. Embedding a two-earner household of the kind [Borella et al. \(2023\)](#) estimate in a model with continuous health capital, health-modulated survival, and endogenous factor prices would compound the curse of dimensionality with the fixed-point problem in prices; to our knowledge no published model carries all four.

The approach here is different, and it is implemented rather than proposed. The agent space carries four types throughout—two education groups for each sex—each with its own measured age-efficiency profile and its own baseline survival schedule, so that the sex- and education-differentiated profiles embed in reduced form the cumulative effects of labor market histories that a structural household model would generate endogenously.

Carrying that heterogeneity from measurement through to welfare is one of this paper’s contributions. Quantitative models of pension reform ordinarily resolve incidence by age and by wealth; resolving it by sex and by education as well requires survival and efficiency schedules estimated separately for each of the four groups, and it changes the answer rather than decorating it. Every experiment below reports welfare and incidence for all four types, and the patterns that result—low-education men the only newborn losers under inaction, low-education women bearing the higher retirement age—are decided by those measured schedules and are unavailable in a model without them. The structural household model remains a natural extension; it is not a prerequisite for the decomposition by sex and education that the paper’s questions require.

Section 3 presents the framework that integrates these strands. It extends the ITED baseline ([Cortés et al., 2024](#)) to general-equilibrium factor pricing, a continuous health capital stock with age-dependent depreciation, and a contribution-rate rule that links demographic aging to the fiscal wedge on formal labor; Sections 4 and 6 then estimate it on Mexican micro-data and price the three pension margins in it, and Section 5 is where the investment assumption of this section meets the Mexican spending data.

3 Model

3.1 Environment and Demographics

Time is discrete. The economy is populated by J overlapping generations indexed by $j = 1, \dots, J$, where $j = 1$ corresponds to age 20 and each model period spans five years. Households enter economic life at $j = 1$, retire exogenously at age j_R , and may live until $j = J$ (age 100).

Agent types. Each individual is indexed by sex $g \in \{m, f\}$ (male, female) and by a permanent skill type $\theta \in \{\theta_L, \theta_H\}$ (low-skilled, high-skilled), where θ_L and θ_H denote the log-productivity intercepts of the two skill groups (see equation (12)). The two skill groups correspond to education groups, and we use the two terms interchangeably. Sex and skill are both drawn at birth and remain fixed over the life cycle. Sex enters the model through two channels: the age-efficiency profile $e_j^{g,\theta}$ that shapes earnings, and the baseline survival schedule $\psi_j^{\circ g,\theta}$ that shapes longevity. Skill enters through three: the log-intercept θ ; the skill-specific component of the age-efficiency profile $e_j^{g,\theta}$, which lets the life-cycle growth of earnings differ across education groups; and the skill-specific component of the baseline survival schedule $\psi_j^{\circ g,\theta}$ in the survival modulation (11), which carries mortality differences across education groups that do not operate through health investment. The health-productivity slope ϱ in (12) is common to both groups. Let $\pi_1^{g,\theta}$ denote the birth share of agents of sex g and skill θ , with $\sum_{g,\theta} \pi_1^{g,\theta} = 1$. The population grows at gross rate $1 + n_p$, so the absolute mass of newborns of type (g, θ) at calendar time t is $\pi_1^{g,\theta} \cdot N_t$.

Idiosyncratic risk. Households face three sources of individual uncertainty: mortality risk, which depends on health; idiosyncratic labor-productivity risk, summarized by an AR(1) shock η ; and health risk, an age-dependent probability of losing a fixed amount of health (Section 3.4.3). Health h is therefore a controlled stochastic state: endogenous through chosen medical spending, risky through the shock. The productivity and health shocks are independent, and there is no aggregate uncertainty.

Individual state. The individual state of a household at the beginning of a period is

$$s = (a, h, \eta), \tag{1}$$

where $a \in \mathcal{A} = [0, \bar{a}]$ are beginning-of-period assets, $h \in \mathcal{H}_h = (0, \bar{h}]$ is health capital, and $\eta \in \mathcal{H}_\eta = \{\eta_1, \dots, \eta_{N_\eta}\}$ is the idiosyncratic productivity component. Sex g , skill type θ and age j are deterministic indices that label the value and policy functions.

Cohort weights. Let $\phi_j^g(\theta)$ denote the stationary mass of age- j households of sex g and skill θ , normalized so that $\sum_g \sum_\theta \sum_{j=1}^J \phi_j^g(\theta) = 1$ with newborn weights $\phi_1^g(\theta) = \pi_1^{g,\theta}/Z$, where

$$Z = \sum_g \sum_\theta \sum_{j=1}^J \pi_1^{g,\theta} \prod_{i=2}^j \left[\bar{\psi}_i^g(\theta) / (1 + n_p) \right]$$

is the normalizing constant, with the convention that the empty product at $j = 1$ equals one. Birth shares are sex- and skill-specific and need not coincide across types. The weights satisfy the forward recursion

$$\phi_{j+1}^g(\theta) = \frac{\bar{\psi}_{j+1}^g(\theta)}{1 + n_p} \phi_j^g(\theta), \quad j = 1, \dots, J-1, \quad (2)$$

where the population-weighted average survival is

$$\bar{\psi}_{j+1}^g(\theta) = \frac{1}{\phi_j^g(\theta)} \int \sum_{k \in \{0,1\}} \Pr(k | j) \psi_{j+1}^{g,\theta}(h'_k(s; j, g, \theta)) d\mu_j^{g,\theta}(s), \quad (3)$$

with $h'_k(s; j, g, \theta)$, $k \in \{0, 1\}$, the two nodes of the health lottery (16) evaluated at the optimal medical spending $m_j(s; g, \theta)$, $\Pr(1 | j) = \pi_j^h$ the shock arrival probability of (17), and $\mu_j^{g,\theta}$ the unconditional age- j population measure for sex g and skill θ , so that $\int d\mu_j^{g,\theta} = \phi_j^g(\theta)$. Terminal-age survival is zero: $\psi_{J+1}^{g,\theta}(\cdot) \equiv 0$.

Within-period timing. The timing within each period is:

1. The household observes its current state (a, h, η) .
2. Working-age households choose (c, ℓ, a', m) . Retired households choose (c, a', m) . Medical spending is committed before the health shock is realized, so investment cannot condition on the shock's arrival: medical spending in the model is a preventive object by construction, building the stock in advance of illness rather than restoring it afterwards.
3. End-of-period assets are a' . Post-investment health is $\tilde{h} = (1 - \delta_j^h) h + H_j(m, h)$, with H_j the deficit-scaled production function (14). The health shock is then realized: with probability π_j^h the household loses ς units of health, and next-period health h' is the realized node of the lottery (16). Survival to next period is realized with probability $\psi_{j+1}^{g,\theta}(h')$, evaluated at the realized h' . Assets of agents who die are voided: they neither enter aggregate capital next period nor flow to survivors. This rules out an accidental-bequest channel and isolates the life-cycle savings motive; the aggregate resource implication is tracked by the void term Λ_{void} introduced in equation (28).

3.2 Preferences

Individuals rank streams of consumption, leisure, and health by expected discounted lifetime utility. The lifetime objective of a newborn household of sex g and type θ is

$$\mathbb{E}_1 \left[\sum_{j=1}^J \beta^{j-1} \left(\prod_{i=2}^j \psi_i^{g,\theta}(h'_{i-1}) \right) u(c_j, \ell_j, h_j) \right], \quad (4)$$

where $\beta \in (0, 1)$, the survival probability $\psi_i^{g,\theta}(h'_{i-1})$ is the probability of being alive at age i given that realized next-period health entering age i is $h'_{i-1} = h_i$, and the expectation is taken over the random path $\{\eta_j\}_{j=1}^J$ of the idiosyncratic shock generated by (13) and over the realizations of the health lottery (16).

The period utility function is a GHH aggregator scaled by a health amenity:

$$u(c, \ell, h) \equiv \Omega(h) \left[\frac{(c - v(\ell))^{1-\gamma} - 1}{1-\gamma} + \chi \right], \quad v(\ell) \equiv \Psi \frac{\ell^{1+\nu_\ell}}{1+\nu_\ell}, \quad (5)$$

$$\Omega(h) \equiv \begin{cases} \exp\left(\Xi \frac{h^{1-\xi} - 1}{1-\xi}\right) & \xi \neq 1, \\ h^\Xi & \xi = 1, \end{cases} \quad (6)$$

with $\Psi > 0$, $\Xi \geq 0$, $\nu_\ell > 0$ (the inverse Frisch elasticity, written with an ℓ subscript to avoid collision with the productivity function ν_j), $\gamma > 0$, and $\chi > -1$. Here $c \geq 0$ is consumption, $\ell \in [0, 1]$ is hours worked, $1 - \ell$ is leisure, and h is the contemporaneous stock of health capital, which simultaneously affects survival, productivity, and the amenity value of consumption. The amenity Ω is nondecreasing in h and strictly positive for every admissible (Ξ, ξ) , and strictly increasing whenever $\Xi > 0$, so health is unambiguously a good wherever the bracket in (5) is positive.

Two conventions fix the health block. The stock is measured as one minus a health deficit, so it is bounded above, and we place its upper endpoint at $\bar{h} = 1$: $h \in \mathcal{H}_h = (0, \bar{h}]$, with $h = \bar{h}$ denoting full health. This choice of units is not free. Health enters labor productivity through ϱh in (12), survival through $\psi_{j+1}^{g,\theta}(h)$ in (11), and its own law of motion (16), so a common rescaling of h would have to be undone simultaneously in all three objects, and the measurement of the stock is what pins the scale down. (The domain is open at the origin because $\Omega(h) \downarrow 0$ as $h \downarrow 0$ whenever $\xi \geq 1$, so $h = 0$ is excluded rather than assigned a utility level.) The normalization $\Omega(\bar{h}) = 1$, by contrast, is free: the amenity multiplies period utility and death carries a continuation value of zero, so replacing Ω by $\lambda\Omega$ for any $\lambda > 0$ leaves preferences unchanged and λ is not identified. The -1 in the exponent of (6) is what imposes $\lambda = 1$; since $(h^{1-\xi} - 1)/(1 - \xi) \rightarrow \log h$ as $\xi \rightarrow 1$, the same subtraction makes the second branch of (6) the continuous limit of the first rather than a separate assumption. With the scale

fixed, Ξ governs the gradient of the amenity in health and ξ its curvature.

The multiplicative amenity is a substantive specification choice, and it is where the utility side of the paper's mechanism lives. Scaling the GHH aggregator by $\Omega(h)$ makes the marginal utility of consumption increasing in health, $u_{ch} = \Omega'(h)(c - v(\ell))^{-\gamma} \geq 0$, with strict inequality whenever $\Xi > 0$, so health and consumption are complements in period utility. The property that matters for behavior, however, is that the marginal rate of substitution between health and consumption, u_h/u_c , is increasing in consumption at every state the household ranks above not being alive: the willingness to pay for health rises with income. Had the amenity entered the composite additively, that ratio would be independent of consumption—health and consumption would be perfect substitutes inside the composite, and the willingness to pay for health would carry no income gradient at all. The multiplicative form therefore does not merely tilt a gradient; it creates one where the additive form has none, and that gradient is the utility-side channel through which income inequality is transmitted into health inequality. Appendix A.1 derives both marginal rates of substitution and records the comparison. Two further properties come with the specification. Because the amenity multiplies the aggregator, $u_c \rightarrow \infty$ as $c \downarrow v(\ell)$, so consumption satisfies an Inada condition that an additive amenity can forfeit; and because $\Omega(h)$ scales u uniformly, it cancels from the intratemporal labor margin, so the wealth-effect-free property of GHH survives on the labor-supply decision.

The constant χ is not a free normalization either. Death carries a continuation value of zero in (4), so the level of u is economically meaningful; and because the -1 in the numerator of (5) itself contributes the constant $1/(\gamma - 1)$ to the bracket, the object that is identified—and that constitutes the flow value of being alive—is the sum $\chi + 1/(\gamma - 1)$ rather than χ alone. That sum determines the household's willingness to pay to reduce mortality risk; states at which the bracket in (5) is negative are states the household ranks below not being alive. Separating χ from $\Omega(h)$ assigns to distinct parameters the two roles the amenity would otherwise play at once— Ξ and ξ govern how utility varies with health, while χ governs the level of utility from being alive at all—and because χ is absent from u_c , it shifts the value function, and hence the value of a statistical life and the demand for survival-improving medical spending, without disturbing the consumption Euler equation or the labor margin.

For retirees ($j \geq j_R$), $\ell_j = 0$ and period utility reduces to $u(c, 0, h) = \Omega(h) \left[\frac{c^{1-\gamma} - 1}{1-\gamma} + \chi \right]$.

3.3 Technology

A representative, competitive firm produces the single consumption good using capital K and aggregate effective labor L according to the Cobb–Douglas technology

$$Y = A K^\alpha L^{1-\alpha}, \quad \alpha \in (0, 1), \quad (7)$$

with $A > 0$ total factor productivity. Capital depreciates at rate $\delta \in (0, 1)$. Constant returns to scale imply that the firm's first-order conditions depend only on the ratio K/L and are therefore invariant to the choice of units, so we write K and L in the stationary per-cohort-mass units of Section 3.6 throughout.

Firm's problem. The firm takes factor prices (r, w) as given and maximizes static profits:

$$\max_{K,L} \{AK^\alpha L^{1-\alpha} - (r + \delta)K - wL\}. \quad (8)$$

The first-order conditions yield

$$r = \alpha A \left(\frac{K}{L} \right)^{\alpha-1} - \delta, \quad (9)$$

$$w = (1 - \alpha) A \left(\frac{K}{L} \right)^\alpha. \quad (10)$$

Factor prices. Both r and w are determined endogenously: given aggregate capital K and aggregate effective labor L , the capital–labor ratio pins down both prices simultaneously through (9)–(10). The interest rate is the object that adjusts to clear the capital market.

3.4 Exogenous Processes

3.4.1 Mortality

Conditional survival probabilities depend on age, sex, skill, and end-of-period health. The probability of surviving from age j to age $j + 1$, realized at the end of age j , is

$$\psi_{j+1}^{g,\theta}(h') = \psi_{j+1}^{\circ g,\theta} \min\{1, \omega + (1 - \omega)h'\}, \quad (11)$$

where h' is the realized next-period health stock determined by the health lottery (16), $\psi_{j+1}^{\circ g,\theta} \in (0, 1]$ is a sex- and education-specific baseline survival schedule—a primitive of the environment, exactly as the sex split is—and $\omega \in (0, 1)$ is the survival floor, the fraction of baseline survival retained as health approaches its lower bound. Under the normalization $\bar{h} = 1$ of Section 3.2, the modulation equals one at full health, so $\psi_{j+1}^{\circ g,\theta}$ is survival at full health; the min caps the modulation at one, so full health cannot raise survival above the baseline schedule. The modulation is piecewise linear in h' : the survival return to health, $(\psi_{j+1}^{g,\theta})'(h') = \psi_{j+1}^{\circ g,\theta}(1 - \omega)$, is constant wherever the cap is slack and zero where it binds, and this derivative is the object that prices survival in the medical-spending condition (56) of Appendix A.2. Terminal-age survival is zero, $\psi_{J+1}^{g,\theta}(\cdot) \equiv 0$, so death by age $j = J$ is certain. Section 4 carries the measurement of the schedules and the floor.

Two mortality channels, one decomposition. Health enters (11) through the term $(1 - \omega)h'$, and that term is the investment channel: medical spending raises next-period health and with it survival, and it is the only survival margin a household controls. The education split of the baseline schedule is a non-investment channel. Education groups differ in mortality for reasons that do not run through the purchase of medical care—differential exposure to occupational hazards, health behaviors, and access to care—and the schedule carries those differences as a measured input, so that the investment channel is not forced to absorb mortality variation it cannot generate. The model's education gap in survival at any age therefore decomposes into a component generated by the education gap in health investment—the baseline schedule multiplied by $(1 - \omega) \Delta_j$ wherever the cap is slack, where Δ_j denotes the education gap in mean next-period health averaged over the population distribution and the shock lottery—and a component inherited from the education gap in the baseline schedules themselves, measured rather than generated. The two components are separately identified by construction, and Section 4 reports the split.

3.4.2 Labor Productivity

Effective labor productivity of a working-age household ($j < j_R$) is

$$v_j(h, \eta; g, \theta) = e_j^{g, \theta} \exp(\theta + \eta + \varrho h), \quad (12)$$

where $e_j^{g, \theta} > 0$ is a deterministic sex- and skill-specific age-efficiency profile, $\theta \in \{\theta_L, \theta_H\}$ is the log-productivity intercept of the skill group, and $\varrho \geq 0$ is the elasticity of productivity with respect to health capital, common to both skill groups. Skill therefore shifts both the level of earnings, through θ , and their life-cycle shape, through $e_j^{g, \theta}$: education groups differ in their returns to experience, and the profile is the object that carries that difference. Health enters symmetrically across groups, through the single slope ϱ ; any skill difference in the level or growth of earnings is attributed to the intercept and the profile, not to a differential return to health. Productivity is zero for retirees ($j \geq j_R$).

The idiosyncratic component η follows the AR(1) process

$$\eta_{j+1} = \rho \eta_j + \varepsilon_{j+1}, \quad \varepsilon_{j+1} \sim \mathcal{N}(0, \sigma_\varepsilon^2), \quad (13)$$

with $|\rho| < 1$.

3.4.3 Health Formation, Health Investment, and Health Risk

Health evolves through three forces: deterministic depreciation at the age-dependent rate $\delta_j^h \in (0, 1)$, repair through chosen medical spending m , and a stochastic health shock. We take the three in turn and close with the deterministic special case that the

specification nests.

Deficit-scaled health production. Medical spending produces health according to

$$H_j(m, h) = \underbrace{\bar{H}_j m^{\zeta_h}}_{\text{remaining deficit}} \frac{\bar{h} - (1 - \delta_j^h) h}{\bar{h}}, \quad \zeta_h \in (0, 1), \quad (14)$$

where $\bar{H}_j > 0$ is an age-dependent scale and the subscript on the curvature ζ_h keeps it distinct from the curvature ξ of the amenity in (6). The return to medical spending is proportional to the health deficit remaining after within-period depreciation: a household already at $\bar{h}$ gains nothing from further spending and one far below it gains a great deal. The deficit is measured after depreciation, so it is the gap actually available to be repaired. Deficit scaling is the clinically natural property of care—treatment restores a loss, and there is less to restore the healthier the patient—and it disciplines the model in a way constant-returns repair cannot; the deterministic limit discussed at the end of this section makes the comparison precise.

Write $\tilde{h} \equiv (1 - \delta_j^h) h + H_j(m, h)$ for post-investment health. Two consequences of (14) follow. First, $\bar{h} - \tilde{h} = (\bar{h} - \bar{H}_j m^{\zeta_h})(\bar{h} - (1 - \delta_j^h) h) / \bar{h}$, so $\tilde{h} < \bar{h}$ for every admissible h if and only if $\bar{H}_j m^{\zeta_h} < \bar{h}$. On that region the upper bound is an asymptote, approached but not attained, and the cap at $\bar{h}$ is retained but slack. The condition restricts the chosen m rather than the parameters, and $\tilde{h}$ is unbounded in m , so the cap can still bind at sufficiently large spending. Second, health production depends on the health stock, so

$$\frac{\partial \tilde{h}}{\partial h} = (1 - \delta_j^h) \left[1 - \frac{\bar{H}_j m^{\zeta_h}}{\bar{h}} \right], \quad \frac{\partial H_j}{\partial m} = \zeta_h \bar{H}_j m^{\zeta_h - 1} \frac{\bar{h} - (1 - \delta_j^h) h}{\bar{h}}, \quad (15)$$

and these are the derivatives that enter the envelope conditions (58) and (60) and the medical-spending condition (56) of Appendix A.2. The marginal product of medical spending falls as health rises.

Stochastic health. Health is risky. With probability π_j^h the household suffers a health shock at age j and loses $\varsigma > 0$ units of health, so next-period health is the two-point random variable

$$h' = \begin{cases} \min\{\tilde{h}, \bar{h}\} & \text{with probability } 1 - \pi_j^h \quad (\text{no shock}), \\ \min\{\max\{\tilde{h} - \varsigma, \underline{h}\}, \bar{h}\} & \text{with probability } \pi_j^h \quad (\text{shock}), \end{cases} \quad (16)$$

with $\underline{h} > 0$ the floor of the shock lottery: an acute loss can displace health below any level reached by depreciation alone, but not below $\underline{h}$. (Health thus lives on $[\underline{h}, \bar{h}]$ once shocks operate; the open lower endpoint of $\mathcal{H}_h$ in (1) is the domain on which

preferences and technology are defined, and $\underline{h}$ is the smallest value the law of motion visits.) We index the two nodes by $k \in \{0, 1\}$, writing h'_k for the realization and $\Pr(k | j)$ for its probability, with $\Pr(1 | j) = \pi_j^h$. The arrival probability inherits its age profile from the depreciation schedule rather than introducing new age-varying parameters,

$$\pi_j^h = \min \left\{ \bar{\pi}^h \frac{\delta_j^h}{\bar{\delta}^h}, \bar{\pi}^{\max} \right\}, \quad \bar{\delta}^h \equiv \frac{1}{j_R - 1} \sum_{i=1}^{j_R-1} \delta_i^h, \quad (17)$$

so that only the scalar $\bar{\pi}^h$, the mean arrival rate over working ages, is free: the ages at which health depreciates fastest are also the ages at which acute losses arrive most often. The cap $\bar{\pi}^{\max}$ prevents the oldest bands, where δ_j^h is largest, from degenerating into certain arrival. The cap has a consequence: wherever it binds, incidence is a constant rather than a profile inherited from δ_j^h , and once every working-age band has reached the cap, the working-age mean incidence ceases to respond to $\bar{\pi}^h$ at all (with δ_j^h increasing in age, the youngest band is the last to cap). The health block carries exactly two risk parameters, $\bar{\pi}^h$ and ς .

Incidence is exogenous, and this is essential. π_j^h and ς depend on age alone, never on income, assets, or skill. Exogenous incidence is the identifying restriction that makes the exercise meaningful. Rich and poor households face the same risk of falling ill; they differ only in what they can do about it. Any health–income gradient the model produces is therefore earned through the response to health shocks rather than assumed through their arrival. Allowing incidence to depend on income would place the paper’s conclusion among its premises. The same discipline governs mortality: the baseline schedule in (11) is fixed at birth with sex and skill, so within a type the only income-varying survival object is health itself.

The deterministic limit. Setting $\bar{\pi}^h = 0$ makes the lottery (16) degenerate at $\min\{\tilde{h}, \bar{h}\}$: deficit-scaled repair without risk. Deleting in addition the deficit factor from (14) recovers the constant-returns rule

$$h' = \min \left\{ (1 - \delta_j^h) h + \bar{H}_j m^{\varsigma_h}, \bar{h} \right\}, \quad (18)$$

familiar from deterministic health-capital models. The baseline is not that rule. Under (18) the return to medical spending is independent of the health stock, so restoring health from just below the ceiling is exactly as productive as restoring it from far below. Whether that matters is an empirical question about the size of depreciation at working ages, and Section 5.2 answers it. The two departures from the rule are complements: deficit scaling makes the marginal product of care fall with health, so heterogeneous households stop short of the ceiling at heterogeneous points, and the shock keeps the cross-section from being reabsorbed into it.

3.5 Government

The government runs two separate balance sheets: a general fiscal block of taxes, public goods and one-period debt, and a self-financed pay-as-you-go (PAYG) pension block. Pension contributions τ^p enter the general budget only indirectly, through the labor-tax wedge faced by workers; they are not a revenue item for the general government.

3.5.1 Fiscal Policy

The government taxes consumption at rate τ^c , labor income at rate τ^ω , capital income at rate τ^k , and medical spending at rate τ^m . It purchases public goods $G = g_y Y$ and may issue one-period debt B . All aggregate quantities (C, K, L, M, Y, G, B) are in the stationary per-cohort-mass units of Section 3.6; C, L, M, Y, G are per-period flows and K, B are stocks. The flow budget is

$$\tau^c C + \tau^m M + \tau^\omega w L + \tau^k r K + B' = G + (1 + r)B, \quad (19)$$

where C is aggregate consumption, K the aggregate capital stock, M aggregate health investment, and B' next-period government debt.

Fiscal closure. $(\tau^c, \tau^\omega, \tau^k, \tau^m, g_y)$ are exogenous policy parameters. In the stationary equilibrium $B' = B$, so debt service rB equals the primary surplus:

$$r B = \tau^c C + \tau^\omega w L + \tau^k r K + \tau^m M - G. \quad (20)$$

B is the residual that closes the budget.

3.5.2 Pension System

A PAYG pension is financed by a payroll tax τ^p on workers' labor income. The system balances each period:

$$\tau^p w L = \overline{pen} N^R, \quad (21)$$

where $N^R = \sum_g \sum_\theta \sum_{j=j_R}^J \phi_j^g(\theta)$ is the mass of retirees. The pension benefit is the replacement-rate parameter κ times average per-worker labor income:

$$\overline{pen} = \kappa \cdot \frac{wL}{N^W}, \quad (22)$$

where $N^W = \sum_g \sum_\theta \sum_{j=1}^{j_R-1} \phi_j^g(\theta)$ is the mass of working-age individuals.

Pension closure. The replacement rate κ is an exogenous policy parameter and the contribution rate τ^p adjusts endogenously to balance the system each period. Substi-

tuting (22) into (21) yields

$$\tau^p = \kappa \frac{N^R}{N^W}, \quad (23)$$

so the contribution rate is the replacement rate scaled by the system dependency ratio N^R/N^W . This pins τ^p down once the stationary distribution is known, and it is the link through which demographic change is transmitted to the tax wedge on formal labor.

3.6 Markets

Throughout, we work in *stationary per-cohort-mass* units. Aggregates are integrals of individual quantities against the stationary population measure $\mu_j^{g,\theta}$, with $\sum_g \sum_\theta \sum_j \phi_j^g(\theta) = 1$. Population growth n_p is already absorbed into the cohort-weight recursion (2), so the aggregates defined below do not grow with the actual population. A symbol such as K in the Markets and Equilibrium sections refers to the stationary capital stock per unit of cohort mass, related to the absolute capital stock by $K_t^{\text{abs}} = K \cdot N_t$, where N_t is total population at t growing at gross rate $1 + n_p$. Stocks and per-period flows are distinguished throughout. *Stocks* (units: goods, or units of agent mass): $K, A_{\text{dom}}, B, N^W, N^R$. *Flows* (units: goods per period, or efficiency units of labor per period): $Y, C, M, G, (n_p + \delta)K, L, \Lambda_{\text{void}}, \overline{p\bar{e}n}$. Gross investment in capital is $(n_p + \delta)K$ per cohort mass per period: δK replaces depreciation and $n_p K$ equips the next cohort, which enters at gross rate $1 + n_p$. The goods-market identity (27) is a flow identity, and every term in it has units of goods per period.

Asset markets are incomplete. Households save in a single non-contingent asset a and cannot borrow, so

$$a' \geq 0. \quad (24)$$

Both factor prices (r, w) are determined endogenously by (9)–(10). Markets clear when the following conditions hold.

Labor market.

$$L = \sum_g \sum_\theta \sum_{j=1}^{j_R-1} \int v_j(h, \eta; g, \theta) \ell_j(s; g, \theta) d\mu_j^{g,\theta}(s). \quad (25)$$

Capital market. Domestic household savings finance the entire capital stock and government debt:

$$K = A_{\text{dom}} - B, \quad A_{\text{dom}} = \sum_g \sum_\theta \sum_{j=1}^J \int a d\mu_j^{g,\theta}(s). \quad (26)$$

This condition, together with the firm's first-order condition (9), determines the equilibrium interest rate.

Goods market.

$$Y = C + M + (n_p + \delta)K + G + \Lambda_{\text{void}}, \quad (27)$$

where every term is a flow in stationary per-cohort-mass units, $(n_p + \delta)K$ is gross investment, and the voided-asset flow is

$$\Lambda_{\text{void}} \equiv \sum_g \sum_{\theta} \sum_{j=1}^J (1 - \bar{\psi}_{j+1}^g(\theta)) \int a'_j(s; g, \theta) d\mu_j^{g,\theta}(s). \quad (28)$$

Government bond market. The government issues exactly B bonds, and households' aggregate asset holdings finance both K and B :

$$A_{\text{dom}} = K + B, \quad (29)$$

which is (26) rearranged.

3.7 Optimization Problems

3.7.1 Net Labor Income

Net labor income at age j is

$$y_j(h, \eta, \ell; g, \theta) = w v_j(h, \eta; g, \theta) \ell (1 - \tau^w - \tau^p), \quad (30)$$

where τ^w is the labor income tax and τ^p the pension contribution rate. All workers in the baseline model are formal; a formalization counterfactual is left for future work.

3.7.2 Workers' Problem

Individuals observe their productivity shock and then choose consumption, savings, labor supply, and health investment. Given labor supply ℓ , household resources at age j are

$$x_j(a, h, \eta, \ell; g, \theta) = (1 + r(1 - \tau^k))a + y_j(h, \eta, \ell; g, \theta). \quad (31)$$

The worker's Bellman equation is

$$V_j(a, h, \eta; g, \theta) = \max_{\substack{c \geq 0, a' \geq 0 \\ \ell \in [0, 1], m \geq 0}} \left\{ u(c, \ell, h) + \beta \sum_{k \in \{0, 1\}} \Pr(k | j) \psi_{j+1}^{g, \theta}(h'_k) \mathbb{E}[V_{j+1}(a', h'_k, \eta'; g, \theta) | \eta] \right\} \quad (32)$$

subject to

$$(1 + \tau^c) c + (1 + \tau^m) m + a' = x_j(a, h, \eta, \ell; g, \theta), \quad (33)$$

$$h'_k \text{ given by (16) with } \tilde{h} = (1 - \delta_j^h) h + H_j(m, h), \quad (34)$$

$$a' \geq 0, \quad m \geq 0, \quad \ell \in [0, 1]. \quad (35)$$

The survival factor sits inside the sum over shock nodes, because a household struck by a shock faces both a lower continuation value and a lower probability of reaching it; collapsing the two would price survival at the wrong health level. The expectation over η' is nested inside the expectation over health and the two are independent by construction, so health shocks carry no information about future earnings beyond their effect on the health state itself. Denote the optimal savings policy by $a'_j(a, h, \eta; g, \theta)$, and analogously for c_j, ℓ_j, m_j .

3.7.3 Retiree's Problem

For retirees ($j \geq j_R$) labor income is zero, so resources are

$$x_j(a; g, \theta) = (1 + r(1 - \tau^k))a + \overline{pen}. \quad (36)$$

The productivity shock η does not enter retirees' resources but is retained as a state to keep the dimension of the state space uniform across ages. The Bellman equation is

$$V_j(a, h, \eta; g, \theta) = \max_{c \geq 0, a' \geq 0, m \geq 0} \left\{ u(c, 0, h) + \beta \sum_{k \in \{0, 1\}} \Pr(k | j) \psi_{j+1}^{g, \theta}(h'_k) \mathbb{E}[V_{j+1}(a', h'_k, \eta'; g, \theta) | \eta] \right\} \quad (37)$$

subject to

$$(1 + \tau^c) c + (1 + \tau^m) m + a' = x_j(a; g, \theta), \quad (38)$$

$$h'_k \text{ given by (16) with } \tilde{h} = (1 - \delta_j^h) h + H_j(m, h), \quad (39)$$

$$a' \geq 0, \quad m \geq 0. \quad (40)$$

At the terminal age $j = J$, $\psi_{J+1}^{g, \theta}(\cdot) \equiv 0$, the continuation value vanishes, and the optimum is $a'_J = 0$ and $m_J = 0$: neither assets nor health carry any continuation value once survival is impossible.

Optimality. The first-order, envelope, and Kuhn–Tucker conditions of the two problems are collected in Appendix A.2; three of their properties organize the economics. First, medical spending earns two returns that the model keeps distinct: a survival return—spending raises next-period health, and with it the probability of reaching the continuation value at all—and a health-capital return—conditional on survival, the extra health that spending produces is valued at the marginal continuation value

of health, which collects its productivity, amenity, and future-survival returns. The medical-spending condition (56) prices both, averaged over the two nodes of the health lottery. Second, survival enters the consumption Euler equation (61) as an effective discount factor, so households that invest in health also lengthen the horizon over which they save; and because next period's marginal utility is evaluated at the realized health node, the health shock is marginal-utility risk against which precautionary saving is the only available insurance—there is no insurance market for the shock. Third, the amenity cancels from the intratemporal consumption–leisure condition (62), so health raises labor supply only through productivity v_j and the labor margin remains free of wealth effects.

3.8 Stationary Competitive Equilibrium

Let $\mu_j^{g,\theta}$ denote the time-invariant population measure of age- j households of sex g and skill θ over the state space $\mathcal{S} = \mathcal{A} \times \mathcal{H}_h \times \mathcal{H}_\eta$, with $\mathcal{A} = [0, \bar{a}]$, $\mathcal{H}_h = (0, \bar{h}]$, and $\mathcal{H}_\eta = \{\eta_1, \dots, \eta_{N_\eta}\}$. The measure is *unconditional*: $\int d\mu_j^{g,\theta} = \phi_j^g(\theta)$, so aggregates are integrals over it without an outer cohort weight. Let $\Pi^\eta(\eta, \eta')$ denote the transition kernel of the AR(1) process (13) discretized to the finite grid $\mathcal{H}_\eta$; the discretization is described in Appendix B.

3.8.1 Aggregate Objects

$$C = \sum_g \sum_\theta \sum_{j=1}^J \int c_j(s; g, \theta) d\mu_j^{g,\theta}(s), \quad (41)$$

$$M = \sum_g \sum_\theta \sum_{j=1}^J \int m_j(s; g, \theta) d\mu_j^{g,\theta}(s), \quad (42)$$

$$L = \sum_g \sum_\theta \sum_{j=1}^{j_R-1} \int v_j(h, \eta; g, \theta) \ell_j(s; g, \theta) d\mu_j^{g,\theta}(s), \quad (43)$$

$$A_{\text{dom}} = \sum_g \sum_\theta \sum_{j=1}^J \int a d\mu_j^{g,\theta}(s), \quad (44)$$

$$N^W = \sum_g \sum_\theta \sum_{j=1}^{j_R-1} \phi_j^g(\theta), \quad N^R = \sum_g \sum_\theta \sum_{j=j_R}^J \phi_j^g(\theta), \quad (45)$$

$$\Lambda_{\text{void}} = \sum_g \sum_\theta \sum_{j=1}^J (1 - \bar{\psi}_{j+1}^g(\theta)) \int a'_j(s; g, \theta) d\mu_j^{g,\theta}(s). \quad (46)$$

3.8.2 Law of Motion for the Distribution

For each measurable set $\mathcal{B} \subseteq \mathcal{S}$, each age $j = 1, \dots, J-1$, each sex g , and each skill θ :

$$\begin{aligned} \mu_{j+1}^{g,\theta}(\mathcal{B}) &= \frac{1}{1+n_p} \int \sum_{k \in \{0,1\}} \Pr(k | j) \psi_{j+1}^{g,\theta}(h'_k(s; j, g, \theta)) \\ &\left[\sum_{\eta' \in \mathcal{H}_\eta} \Pi^\eta(\eta, \eta') \mathbf{1}[(a'_j(s; g, \theta), h'_k(s; j, g, \theta), \eta') \in \mathcal{B}] \right] d\mu_j^{g,\theta}(s), \end{aligned} \quad (47)$$

where $h'_k(s; j, g, \theta)$, $k \in \{0, 1\}$, are the two nodes of the health lottery (16) evaluated at the optimal policies. Mass flows to both nodes in proportion to the arrival probability, each node surviving at its own health level, so the health shock is a source of within-cohort dispersion in the stationary distribution and not only of individual risk. The factor $1/(1+n_p)$ expresses the unconditional next-period measure in per-newborn-cohort units, consistent with the cohort-weight recursion (2). The recursion runs for $j = 1, \dots, J-1$; no measure $\mu_{J+1}^{g,\theta}$ is defined, because $\psi_{J+1}^g \equiv 0$ implies that all age- J agents die at the period boundary. The initial condition for newborns is

$$\mu_1^{g,\theta}(\mathcal{B}) = \phi_1^g(\theta) \cdot \sum_{\eta_1 \in \mathcal{H}_\eta} \pi_{\text{erg}}^\eta(\eta_1) \mathbf{1}[(0, h_0, \eta_1) \in \mathcal{B}], \quad (48)$$

where π_{erg}^η is the ergodic distribution of Π^η and $h_0 \in (0, \bar{h}]$ is the initial health endowment, a parameter of the model. Newborns hold no assets and draw η from the ergodic distribution, so all initial heterogeneity across households of a given type is in the productivity draw.

3.8.3 Definition

Definition 1 (Stationary Recursive Competitive Equilibrium). *Given exogenous policy $(\tau^c, \tau^\omega, \tau^k, \tau^m, g_y, \kappa)$, birth shares $\{\pi_1^{g,\theta}\}_{g,\theta}$, population growth n_p , initial health endowment $h_0 \in (0, \bar{h}]$, and parameters of preferences, technology, mortality, productivity, and health formation, a stationary recursive competitive equilibrium consists of*

1. *value functions $\{V_j(\cdot; g, \theta)\}_{j=1,\dots,J;g,\theta}$ and policy functions $\{c_j, \ell_j, m_j, a'_j\}(\cdot; g, \theta)$ defined on $\mathcal{S}$;*
2. *factor prices (r, w) ;*
3. *endogenous fiscal and pension variables $(B, \tau^p, \overline{pen})$;*
4. *aggregates $(K, L, C, M, A_{\text{dom}}, N^W, N^R, \Lambda_{\text{void}})$ in stationary per-cohort-mass units (Section 3.6);*
5. *cohort weights $\{\phi_j^g(\theta)\}_{j,g,\theta}$ and population-average survivals $\{\bar{\psi}_{j+1}^g(\theta)\}_{j,g,\theta}$;*

6. stationary distributions $\{\mu_j^{g,\theta}\}_{j,g,\theta}$;

such that:

1. Worker optimality. For each (g, θ) and $j < j_R$, V_j solves the worker's Bellman problem (32)–(35), with policy functions (c_j, ℓ_j, m_j, a'_j) .
2. Retiree optimality. For each (g, θ) and $j \geq j_R$, V_j solves the retiree's Bellman problem (37)–(40), with policy functions (c_j, m_j, a'_j) and $\ell_j \equiv 0$.
3. Firm optimality. (r, w) satisfy (9)–(10) given (K, L) .
4. Government budget. B satisfies (20).
5. Pension balance. τ^p satisfies (23) and $\overline{pen}$ satisfies (22).
6. Market clearing. Labor (25), capital (26), and goods (27).
7. Cohort weights. $\phi_j^g(\theta)$ satisfy (2) with $\bar{\psi}_{j+1}^g(\theta)$ given by (3) and birth shares $\phi_1^g(\theta) \propto \pi_1^{g,\theta}$, normalized so that $\sum_g \sum_\theta \sum_j \phi_j^g(\theta) = 1$.
8. Stationarity of the distribution. For each (g, θ, j) , $\mu_j^{g,\theta}$ satisfies the law of motion (47) with the newborn condition (48).
9. Aggregate consistency. $(C, M, L, A_{\text{dom}}, N^W, N^R, \Lambda_{\text{void}})$ are computed from (41)–(46), all in stationary per-cohort-mass units.
10. Demographic–distributional fixed point. Conditions 7 and 8 are mutually dependent. The population-average survivals $\bar{\psi}_{j+1}^g(\theta)$ are functionals of the stationary distribution $\mu_j^{g,\theta}$ through (3), while $\mu_j^{g,\theta}$ depends on $\bar{\psi}_{j+1}^g(\theta)$ through the $1/(1 + n_p)$ scaling in (47) that anchors the cohort weights. Equilibrium requires $(\{\phi_j^g(\theta)\}, \{\bar{\psi}_{j+1}^g(\theta)\}, \{\mu_j^{g,\theta}\})$ to be a joint fixed point of (2), (3), and (47), evaluated at the equilibrium policy functions.

Condition 10 is what distinguishes this equilibrium from one with exogenous mortality. Because survival depends on a state that households choose, the age distribution of the population is an equilibrium object rather than a demographic input, and the same medical-spending decisions that determine health also determine the dependency ratio that prices formal labor.

3.9 Existence and Computation

We do not prove existence. The standard arguments for stationary equilibria in this class take the survival schedule as a primitive; here it is a functional of the equilibrium distribution through (3), and the household objective contains the product $\psi_{j+1}^g(h') \mathbb{E}[V_{j+1}]$, whose concavity in (a', m) is not implied by the restrictions placed on

preferences and on H_j . Existence is therefore established constructively: the algorithm converges to a point satisfying every condition of Definition 1, and Appendix B reports the residuals that support the claim. One closure choice bears repeating: assets of decedents are voided rather than bequeathed or annuitized, which isolates the life-cycle saving motive at the cost of a resource leakage, Λ_{void} in (27). An accidental-bequest or annuity closure is the natural alternative and is left to future work.

Two features of the closure organize everything that follows. First, the PAYG block ties the contribution rate to the system dependency ratio, $\tau^p = \kappa N^R/N^W$, so any force that moves the ratio of retirees to workers moves the labor-tax wedge one for one. Second, because survival is endogenous to medical spending, health investment feeds forward into the dependency ratio itself: longer lives raise N^R/N^W , and hence τ^p . The interaction of these two links — demographic pressure transmitted to the fiscal wedge, and health-driven longevity feeding back into that same wedge — is the *fiscal amplification loop*. It is the mechanism the experiments of Section 6 stress from the dependency side, and the reason better population health raises the wedge under this benefit rule: the identity carries no productivity term through which longer lives could pay for themselves.

4 Calibration

We calibrate the model to México in two steps. The first fixes the exogenous processes and the externally assigned parameters: demographics, the sex- and education-specific baseline survival schedules and the survival floor, the sex- and skill-specific age-efficiency profiles and skill intercepts, the productivity process, the health-depreciation schedule, and the fiscal block. The second estimates the remaining nine parameters—preferences, the health-production technology, the health-risk process, and the health-productivity slope—by classical minimum distance on ten moments. Section 4.1 describes the data, Section 4.3 the first-step inputs, Section 4.4 the estimator and its targets, Section 4.5 the estimates and their fit, and Section 4.6 what the calibrated model delivers on moments it was not asked to match.

Three limits on identification govern how the estimates may be read, and we state them where they bind. The health-productivity slope is identified only jointly with the model’s internal health scale. The two health-risk parameters share their identifying moments with the rest of the health block. And the two targeted survival levels overlap a measured first-step input, so they verify the staging of that input rather than supplying independent information about the parameters.

4.1 Data

Two Mexican household surveys carry the calibration, and each covers the part of the life cycle the other does not. Table 1 lists them alongside the official and international

sources that supply the remaining inputs.

The National Survey of Household Income and Expenditure (ENIGH), INEGI's repeated cross-section of household budgets, measures the working life. It supplies the sex- and skill-specific age-efficiency profiles and skill intercepts that shape earnings, the birth composition of the four types, and seven of the ten moments the second step targets: the three medical-spending shares, the age slope of spending, the two income elasticities, and the skill premium. We pool the 2016 and 2018 waves—the years contemporaneous with the health survey, and the era of the Seguro Popular arrangement whose steady state the model represents—and hold the later waves out. All ENIGH statistics use the survey's expansion factors, with standard errors clustered on primary sampling units, and household-level moments are assigned to the age band of the household head.

The Mexican Health and Aging Study (ENASEM, known in its harmonized form as MHAS) measures the years over which mortality operates. It is a panel of adults aged 50 and over, interviewed at roughly three-year intervals, and we use the harmonized waves of 2012, 2015, 2018 and 2021. It supplies the four sex- and education-specific baseline survival schedules above age 50, the survival floor, the age profile of health depreciation, and the two targeted survival levels. Health is measured there as one minus a deficit index built from a reversible subset of the survey's items—functional status, depressive symptoms, and self-rated health—chosen because medical care can plausibly restore them.

Three further sources fill gaps neither survey can. Official CONAPO life tables carry baseline survival below age 50, where no education split of Mexican mortality exists; the 2020 INEGI census fixes the old-age dependency ratio that pins the population growth rate; and CONAPO's projections supply the 2050 dependency ratio the experiments impose. The World Health Organization's Global Health Expenditure Database supplies the out-of-pocket share of Mexican health spending, which converts the model's full resource cost of care into the out-of-pocket quantity ENIGH observes, and the public share of health financing that is netted out of government consumption.

One measurement rule governs every use of the expenditure survey. ENIGH captures a fraction of the out-of-pocket total recorded in the national health accounts, so we use it for shapes, gradients and composition and never for levels: no peso figure from it is reported as a national total, and no result rests on the absolute size of any spending flow. The spending moments are accordingly shares, slopes and elasticities.

The remaining parameters are taken from external evidence rather than estimated here. Capital depreciation and the capital share are Mexico-specific literature values; the persistence and innovation variance of the idiosyncratic productivity process come from the canonical restricted-income-profile literature, time-aggregated to the model's five-year period, because ENIGH is a repeated cross-section and cannot identify earnings dynamics; and the tax rates are effective, concept-matched rates rather than statutory ones. Section 4.3 gives each with its source. The aggregate ratios against

which the calibrated equilibrium is compared—a capital–output ratio of 3.29 and an investment share of 20.2 per cent—are Penn World Table full-sample averages for México, and public pension spending of 4.5 per cent of GDP is the OECD figure for 2022.

Table 1: Data sources and what each supplies.

SOURCE	TYPE	COVERAGE	SUPPLIES
ENIGH (INEGI)	Household budget survey	Pooled 2016, 2018	Age-efficiency profiles; skill intercepts; birth shares; seven targeted moments
ENASEM/MHAS	Panel, ages 50+	Harmonized 2012, 2015, 2018, 2021	Baseline survival by sex and education; survival floor; health depreciation; two targeted survival levels
CONAPO	Official life tables and projections	Life tables; 2050 projection	Baseline survival below age 50; the 2050 dependency ratio
INEGI Censo	Population census	2020	Old-age dependency ratio, which pins population growth
WHO GHED	Health accounts database	2020	Out-of-pocket share of health spending; public health financing share

Notes. Each survey covers the segment of the life cycle the other does not: ENIGH the working years, ENASEM the years over which mortality operates. Externally assigned parameters and the aggregate comparison ratios are listed with their sources in Section 4.3.

4.2 Units and Normalization

The model is dimensionless until two normalizations fix its scale, and every quantitative statement in the paper depends on stating them once and applying them consistently.

Time. A model period is five years; $j = 1$ is age 20 and retirement is at $j_R = 10$ (age 65). Output, consumption, medical spending, and the wage bill are five-year flows; capital and government debt are stocks. Ratios of a stock to a flow are therefore reported at annual frequency by multiplying the model (five-year) ratio by five: the annual capital–output ratio is $5(K/Y)$ and the annual debt–output ratio is $5(B/Y)$. Rates are annualized geometrically, $r^{\text{ann}} = (1 + r)^{1/5} - 1$ for the five-year return r .

Labour. Effective labour is measured in efficiency units normalized so that the age-efficiency of a male aged 20–24 equals one; the wage w is the price of one such efficiency unit.

Population. All aggregates are in stationary per-cohort-mass units (Section 3.6); levels therefore have no independent meaning across calibrations and only ratios, rates, and within-calibration comparisons are interpreted.

4.3 First-Step Inputs

Table 2 collects the externally assigned and first-step parameters with their values and sources. Every aggregate parameter is sourced for México at a stated concept—effective tax rates rather than statutory ones, government consumption net of public health, a dependency-matched population growth rate—and this subsection records how the data discipline each block.

Table 2: Externally assigned and first-step parameters (México calibration).

DESCRIPTION		VALUE	TARGET/SOURCE
Demographics and technology			
J, j_R	Period length	5 years	model frequency
n_p	Periods of life; retirement period	17, 10	max age 100; retire at 65
α	Population growth	3.51%/yr (18.8% per 5-yr)	dependency-matched: $65+/(20-64) = 0.1414$, INEGI Censo 2020 (see text)
δ	Capital share	0.40	México factor shares (see text)
A	Capital depreciation	5%/yr (22.6% per 5-yr)	México-specific literature value (see text)
	TFP (scale)	1.60	normalization, conditional (see text)
Preferences (fixed externally)			
γ	GHH curvature	2.0	—
ν_t	Inverse Frisch	2.0	—
β	Discount factor, per 5-yr	0.998 ⁵	external; K/Y -invariant (see text)
Fiscal and pension			
τ^c, τ^w, τ^k	Consumption, labour, capital taxes	0.09, 0.10, 0.12	effective, concept-matched rates (see text)
τ^m, g_y	Medical tax (baseline); G/Y	0, 0.090	g_y ex public health (see text)
κ	Pension replacement rate	0.40	flat pension ÷ average earnings (see text)
Labour (ENIGH 2016, 2018; this paper)			
θ_L, θ_H	Skill intercepts, log points	-0.020, +0.020	ages 20–24; earnings-based, divided by $1 + 1/\nu_t$
$e_j^{g,\theta}$	Age-efficiency profiles, efficiency units	profiles	sex- and skill-specific; male 20–24 $\equiv 1$
$\pi_1^{g,\theta}$	Birth shares	shares	sex-by-education composition at age 20
Productivity and health			
ρ, σ_ε	AR(1) of η , per 5-yr	0.837, 0.284	external (see text)
δ_j^h	Health depreciation, per 5-yr	0.0011 $\rightarrow$ 0.186	age-rising; ENASEM (see text)
$\psi_j^{g,\theta}$	Baseline survival, sex- and education-specific	schedules	ENASEM + CONAPO (see text)
ω	Survival floor as $h \downarrow 0$	0.623	ENASEM
h_0	Initial health, health units	$\bar{h} = 1$	—
Measurement			
s	Out-of-pocket share of health spending	0.41	WHO GHED; see Section 4.4

Notes. Rates with “/yr” are annual; per-5-yr values are the model-period magnitudes used in computation. Tax rates are effective, concept-matched rates, not statutory rates (see text). τ^p is not listed because it is endogenous (it clears the PAYG block). The internally estimated parameters— $\Psi, \Xi, \bar{H}_0, h^{\text{slope}}, \zeta_h, \varrho, \chi, \bar{\pi}^h, \varsigma$ —are listed in Table 4. The δ_j^h range spans the youngest working band to the terminal bands (flat from age 90); its smallness at working ages is the fact behind the health-block design discussed in Section 5.2.

Technology and demographic scale. Capital depreciates at $\delta = 0.05$ per year, the México-specific value of [Bergoeing et al. \(2002\)](#), who validate it against the UN-reported share of capital consumption in Mexican GDP and consider and reject 0.08; the capital share $\alpha = 0.40$ follows the factor-share measurement of [García-Verdú \(2005\)](#). The TFP level $A = 1.60$ is a normalization, and one caveat attaches to it: the model is non-homothetic, because health is bounded and the felicity carries a flow value of being alive, so output scale is not neutral and the estimates below are conditional on this value. Population growth is set so that the model’s stationary age distribution reproduces the 2020 census: $n_p = 3.51\%$ per year delivers an old-age dependency ratio of 0.1414 under the model’s own survival schedules. This is a high growth rate relative to what México experienced over the decades that produced the 2020 age pyramid, and that is the price of representing a 2020 age distribution with a stationary demography. Because the required n_p depends on the survival schedules, it is re-solved by bisection whenever those change, so the dependency ratio is matched by construction rather than approximately, and it is accordingly not among the checks of Section 4.6. The discount factor is fixed externally at $\beta = 0.998^5$. It could in principle be disciplined by the capital–output ratio, but that ratio is an equilibrium object while estimation is

carried out in partial equilibrium, and the capital stock is in any case insensitive to it, because the interest-rate response stabilises capital and because the effective discount is β times survival, which dominates at the ages where wealth is held.

Mortality. The four baseline survival schedules $\psi_j^{g,\theta}$, one per sex-by-education group, are estimated from the ENASEM panel: deaths counted between waves identify survival at ages 50 and above, with the education gradient carried by an education main effect and an education-by-age interaction, which keeps the thin high-education cells at the oldest ages usable. Below age 50 the schedules are spliced to official CONAPO life tables, pooled across education, since no education split of Mexican mortality exists at those ages; the education gradient of baseline survival therefore enters only from age 50. The survival floor $\varpi = 0.623$ —the fraction of baseline survival retained as health approaches its lower bound in the modulation (11)—is estimated from the mortality gradient in the measured deficit index among the same population.

The schedules are capped at one, and measured survival at ages 50–69 is high enough that the cap clips the estimated education gradient there—entirely for women, partially for men—so the input understates the education difference in survival at exactly the ages the targeted survival moments cover. An age-varying floor would un-clip it, and we leave that to future work.

Labour productivity. The labour block is estimated from the pooled ENIGH 2016 and 2018 cross-sections. Taking logs of the productivity equation (12) gives

$$\log w_{i,t} = \underbrace{\log e_{j(i,t)}^{g,\theta}}_{\text{age-by-sex-by-skill profile}} + \underbrace{\theta_i}_{\text{skill intercept}} + \underbrace{\varrho h_{i,t}}_{\text{health-productivity slope}} + \eta_{i,t}, \quad (49)$$

from which we recover the age-efficiency profiles $e_j^{g,\theta}$ and the intercepts θ_L, θ_H , normalized so that the male 20–24 cell equals one. Two features of the block matter downstream. First, the profiles are skill-specific, and the difference between them is stark: the low-skill male profile peaks at ages 45–49 and falls back to roughly its age-20 level by 60–64, while the high-skill male profile roughly doubles by its peak in the fifties. The life-cycle growth of the skill premium is therefore a measured input rather than a model outcome, which is why the earnings objects of Section 4.6 are reported as carried rather than as tests. Second, the profiles are estimated from log earnings but consumed by the model as productivity, and GHH labour supply re-amplifies any productivity shifter into earnings by the factor $1 + 1/\nu_\ell = 1.5$. We therefore divide the estimated log-earnings objects by that factor before they enter the model, so that the model reproduces the measured earnings gaps rather than amplifying them a second time. An hourly-wage profile built from reported hours, under which no amplification is implied at all, is preferable in principle and is left for future work.

The persistent component η would ordinarily come from the residuals of this regression, but ENIGH is a repeated cross-section and is uninformative about earnings dynamics, and no accessible Mexican household panel exists at the model's frequency. We therefore take $(\rho, \sigma_\varepsilon)$ from outside the data. The annual persistence and innovation variance are the average of representative estimates from the restricted-income-profile literature (Storesletten et al., 2004; Guvenen, 2009; Hubbard et al., 1994), $\rho_{\text{ann}} \approx 0.965$ and $\sigma_{\varepsilon, \text{ann}}^2 \approx 0.0185$; time-aggregating the annual process to the five-year period by skip-sampling gives $(\rho, \sigma_\varepsilon) = (0.837, 0.284)$, with the innovation scale re-pinned so that the model's stationary dispersion of log earnings matches that observed for Mexican formal-sector workers (Puggioni et al., 2022). This is the input to which the aggregate results are most sensitive, since the persistence of idiosyncratic income risk is the dominant driver of the calibrated capital stock.

The health–productivity slope ϱ in (49) is estimated inside the model rather than fixed at an external anchor, and two features of the environment force the choice. Working-age health cannot be measured: the Latin American health-and-aging surveys—México's ENASEM, Costa Rica's CRELES, the Panama SABE study—all begin at age 50, so no health measure exists over most of the working life at which the wage slope operates. United States work such as Hosseini et al. (2022) can use a panel with health observed from early adulthood; the region has none. And even where health is observed, an external regression of log wages on the health stock is inconsistent with the model's own data-generating process, since health is chosen through income-elastic medical investment and is therefore correlated with the persistent productivity shock. Estimating ϱ inside the model addresses both problems at the price of an interpretation constraint stated below: the data identify it only jointly with the model's internal health scale.

Health depreciation. Newborns are endowed at the health ceiling, $h_0 = \bar{h} = 1$. Over the ages at which health is observed, the age-rising depreciation rate δ_j^h is disciplined by the reversible deficit index of Section 4.1, and the estimand is the contraction of the health stock among individuals observed in both adjacent waves. Measuring within survivors decouples depreciation from survival at the cost of conditioning on survival, so the resulting profile is a lower bound on the force of health decline: those who drop out are those whose health fell fastest. The three-year wave spacing is rescaled to the model's five-year period, and working-age depreciation—unobserved for want of a working-age panel—is set by log-linear extrapolation of the observed profile downward in age, capped at the highest reliably estimated band. The schedule rises from 0.0011 at ages 20–24 to 0.186 from age 90 on.

The smallness of the working-age values is the binding fact for the health block. At a five-year depreciation of a tenth of one per cent, a constant-returns health technology returns essentially every household that spends at all to full health, and it is that degeneracy which the deficit-scaled, risky specification of Section 3.4.3 exists to break;

the evidence is in Section 5.2.

Fiscal block. Every tax parameter enters at an effective, concept-matched rate. The consumption tax $\tau^c = 0.09$ is the rate implied by Mexican VAT and excise collections relative to private consumption: the OECD’s VAT revenue ratio for México is about 0.35, so the statutory 16% rate overstates the burden households face by roughly half. The labour tax $\tau^\omega = 0.10$ is the income-tax-only wedge on the average worker—the OECD puts it near 9.4% in 2020, and effective estimates net of social contributions at 7.7–8.5% (Bachas et al., 2022). Social-security contributions are excluded from it because the model levies the pension contribution separately and endogenously: τ^p clears the PAYG block through equation (23) and equals 5.66% at the baseline, so the combined labour wedge is about 15.7% against a measured total near 20%, the difference being employer contributions outside the model. Because τ^ω is flat, distributional effects running through the progressivity of the Mexican income tax are outside the model. The capital tax $\tau^k = 0.12$ sits inside the range of effective capital-tax estimates for México (0.108–0.121; Bachas et al., 2022) and is consistent with the marginal rates of Antón Sarabia (2005). The public-goods share $g_y = 0.090$ is government consumption net of public health—12.2% of GDP in 2020 less domestic government health spending of 3.2%—so that the public financing which pays for the non-out-of-pocket share of care is counted once, here and not again in the measurement wedge of Section 4.4. The replacement rate $\kappa = 0.40$ is the concept the model’s flat pension implements, a flat benefit divided by economy-wide average earnings, at the value documented for México by Bonthuis (2024). The stationary debt level B is the residual that closes the general budget through (20); both it and τ^p are equilibrium objects.

4.4 Second-Step Estimation

The parameters that neither external evidence nor the first step pins down are estimated jointly by classical minimum distance. The estimated vector is

$$\Theta_{\text{MD}} = (\Psi, \Xi, \bar{H}_0, h^{\text{slope}}, \zeta_h, \varrho, \chi, \bar{\pi}^h, \varsigma) :$$

the labour-disutility scale Ψ ; the amenity scale Ξ governing how much health raises the utility value of consumption; the scale, age slope, and curvature of the health-production technology,

$$\bar{H}_j = \bar{H}_0 \exp(-h^{\text{slope}} j), \quad \zeta_h \in (0, 1), \quad (50)$$

entering the deficit-scaled production function (14), where a positive h^{slope} tilts the technology downward with age and the sign is estimated rather than imposed; the health–productivity slope ϱ ; the flow value of being alive χ ; and the mean arrival rate $\bar{\pi}^h$ and size ς of the health shock. Two adjacent parameters are not estimated.

The amenity curvature ξ is held fixed at an interior value rather than estimated, and we report it as a convention rather than as an estimate. The discount factor is fixed externally, since the capital–output ratio that would discipline it is an equilibrium object while estimation is carried out in partial equilibrium.

The estimator minimizes the sum of squared proportional deviations of the ten model moments from their data counterparts, so that a ten-per-cent miss is priced identically wherever it occurs. Appendix B gives the objective, the weighting choice and the reason inverse-variance weighting was rejected, and the optimizer.

Targets. Table 3 collects the ten targets. Nine are estimated from Mexican micro-data; the hours target is a conventional calibration value. Each is there for a reason the model can be held to. The hours target fixes the scale of labour disutility. The three medical-spending shares and the age slope of spending pin the level and shape of the age profile of medical care: because a slope is invariant to a common multiplicative shift, the slope constrains the shape while the 65–79 share pins the old-age level and the 50–64 share completes the profile. The two income elasticities discipline how strongly spending responds to income, which is the margin on which the paper’s transmission from income to health operates. The skill premium disciplines the earnings return to health. And the two survival levels tie the model’s mortality at the ages where it can be measured to the counted deaths that produced the first-step schedules.

Table 3: Targeted moments, values, and sources.

TARGETED MOMENT	TARGET	SE	SOURCE
Labour			
Prime-age male hours, share of time endowment	0.333	—	convention (see text)
Medical spending			
Income elasticity of medical spending, pooled	0.575	0.029	ENIGH 2016–2018
Income elasticity of medical spending, within age	0.573	0.029	ENIGH 2016–2018
Medical-spending share of expenditure, ages 25–34	0.0182	0.0004	ENIGH 2016–2018
Medical-spending share of expenditure, ages 50–64	0.0231	0.0004	ENIGH 2016–2018
Medical-spending share of expenditure, ages 65–79	0.0332	0.0009	ENIGH 2016–2018
Log-slope of mean medical spending, ages 25–79	0.0675	0.0082	ENIGH 2016–2018
Earnings			
Skill premium in log earnings, ages 25–34	0.659	0.013	ENIGH 2016–2018
Survival			
Five-year survival, low education, ages 50–64	0.951	0.004	ENASEM
Five-year survival, high education, ages 50–64	0.972	0.006	ENASEM

Notes. Survey standard errors are cluster-robust on primary sampling units (the log-slope’s is a PSU-clustered bootstrap; the survival moments’ are delta-method on clustered wave-to-wave survival). The hours target is a convention—a forty-hour week as one third of the time endowment—with no producing estimate behind it; it disciplines the labour-disutility scale and is not an estimated moment, which is why its standard-error cell is empty. No moment pairs one-to-one with one parameter; identification is discussed in the text.

Measurement. Two decisions connect what the survey records to what the model computes, and both change the estimand rather than merely the arithmetic.

The first is a wedge between the two spending concepts. ENIGH records out-of-pocket spending, while the model’s m is the full resource cost of care. The model side of every spending moment therefore records $s \cdot m$, with $s = 0.41$ the out-of-pocket share of Mexican health spending. This is a recording convention and not a price: the household pays the full m in its budget, and the public financing of the remainder sits outside the model—removed once, from government consumption, in Section 4.3.

The second is the estimand for the two elasticities. These are Poisson pseudo-maximum-likelihood regressions of medical spending in levels on log labour income, with wave fixed effects, on working-age heads with positive labour income and with zero-spending households retained. The familiar alternative, a log-log regression conditional on positive spending, drops nearly half of the sample frame, while the model analogue drops essentially nothing: the two would be different objects sharing one name. The Poisson estimand delivers the elasticity of the conditional mean with the zeros in, which is what the model computes. Income is labour income, matching the model’s $w \vee \ell$, and the sample is restricted to working ages, since retirees face a constant pension and carry no income variation for the model to match. The within-age

version repeats the regression band by band and averages the slopes by population mass. The spending shares are expansion-weighted means of health spending over monetary expenditure, with the model analogue $(1 + \tau^m)m / [(1 + \tau^c)c + (1 + \tau^m)m]$; the skill premium is the high-minus-low gap in log labour earnings among persons 25–34, with the model side computed as the exact counterpart of the same regression, so that the two cannot drift apart in construction.² The survival moments are counted deaths between ENASEM waves for persons 50–64, converted to the model’s five-year frequency and split by completed education. One measurement choice attaches to them: the model’s average survival modulation is anchored to the modulation evaluated at the mean measured deficit index, which is the closest this calibration comes to a mapping between the model’s health units and a measured index. It is routed through a first-step input rather than through any comparison of levels.

The two survival moments have a different status from the rest of the table. Since the education split of survival is a measured first-step input, these two levels overlap that input, and what they verify is that the schedules, the health modulation, and the population composition combine to reproduce the levels the schedules were estimated from. They are retained for that verification. The overlap is not an identity—the input comes from an unweighted estimator while the targets are expansion-weighted, and the two differ by about ten per cent on the 50–64 education gap—but the education survival *gap* is treated as a diagnostic rather than a target, and reported in Section 4.6. One feature of these moments bounds what they can show: the survey’s high-education share at 50–64 is about 20%, while the model is stationary, cannot represent México’s education expansion, and carries over 50% at every age.

No second-moment spending target appears in the table. Two candidates were constructed—the cross-sectional dispersion of medical spending and the wave-to-wave persistence of out-of-pocket spending—and neither is an object this model can match, for reasons given in Section 4.6. The consequence for identification is that the two health-risk parameters have no dedicated identifying moment and are disciplined jointly by the moments that respond to them, chiefly the young and middle-aged spending shares and the survival levels.

Identification. No moment pairs one-to-one with one parameter, and the division of labour is economic rather than mechanical. The hours target disciplines the labour-disutility scale. The spending shares, the age slope, and the two elasticities jointly discipline the three parameters of the health technology and, through the level of spending they imply, the amenity scale and the flow value of being alive, which has no assigned moment and is disciplined indirectly through the spending it induces; estimation additionally excludes, by a smooth penalty, the region in which any household

²Both earnings moments condition on positive earnings in the data, while the model has no participation margin and averages over all states. Participation differs by education and sex, and the resulting selection is not corrected.

would assign negative value to survival. The skill premium disciplines the health–productivity slope jointly with the health block, which determines the health gap that slope multiplies. And the survival levels, together with the spending profile, discipline the two shock parameters.

Three constraints on interpretation follow. First, the earnings data identify only the product of the health–productivity slope with the health gap between education groups, and the scale of health is fixed by the model’s own normalization rather than by any measurement. The estimate of ϱ is therefore a structural parameter meaningful relative to the model’s health units and is not comparable to regression coefficients in the health-economics literature; it is legitimate for counterfactuals, which is what it is for. Second, the model’s education gap in survival decomposes, as in Section 3.4.1, into a component inherited from the measured baseline schedules and a component generated by the education gap in health investment. Only the second is the mechanism, so we report the decomposition rather than the total gap. Third, a parameter estimated at a boundary of its admissible set is a diagnosis about the objective rather than a measurement of the parameter, and two of the nine estimates sit at boundaries.

Estimation in partial equilibrium. Parameters are obtained by solving the household problem at fixed anchor prices rather than by clearing markets at every candidate parameter vector. The device concerns how the estimates are computed, not the model: the counterfactuals of Section 6 are solved in general equilibrium with factor prices and the contribution rate responding, which is essential rather than incidental, since the loop the paper quantifies runs through exactly those prices. After estimation, the model’s general equilibrium is solved at the estimates—including the re-bisection of population growth to the dependency target under the resulting survival schedules—and every number in the next two subsections is evaluated at that equilibrium’s own prices and matched demographics, which is the configuration the experiments run in. The anchor and the resulting equilibrium are not identical. Re-evaluating the estimates at the equilibrium’s own prices improves the objective relative to the estimation anchor rather than degrading it. Re-estimating with the market-clearing loop inside the objective remains the natural robustness check.

4.5 Calibration Results

Table 4 reports the nine parameter values and Table 5 the fit of the ten targeted moments, both evaluated at the baseline’s own general-equilibrium prices with population growth re-bisected to the dependency target. Standard errors are not reported, for the reasons given in Appendix B. Two of the nine parameters sit at boundaries of their admissible sets and are read below as diagnoses rather than as measurements.

Table 4: Minimum-distance parameter vector $\hat{\Theta}_{\text{MD}}$.

DESCRIPTION		VALUE
Preferences and labour		
Ψ	Labour-disutility scale	24.5
Ξ	Health-amenity scale	0.0017 [†]
χ	Flow value of being alive	29.2
Health technology		
$\bar{H}_0$	Health-production scale	3.36
h^{slope}	Age slope of health production	-0.0188
ζ_h	Health-production curvature	0.489
Health risk		
$\bar{\pi}^h$	Mean shock arrival rate	0.0100 [†]
ς	Shock size	0.283
Earnings		
ϱ	Health-productivity slope	0.983
ξ	Amenity curvature	0.50 (pinned)

Notes. No standard errors are reported (Appendix B). [†]At a lower bound of the admissible set. Boundary values are diagnoses about the objective rather than measurements, and are read as such in the text. The amenity curvature ξ is held at an interior value and reported as a convention: any admissible value delivers the same fit.

The interior estimates. Medical care is productive but sharply diminishing: the health-production scale is large while the curvature stays near one half ($\bar{H}_0 = 3.36$, $\zeta_h = 0.489$), which is the configuration that lets heterogeneous households stop short of the health ceiling at heterogeneous points. The age profile of the technology is estimated to rise rather than fall with age. In the convention of (50), $\hat{h}^{\text{slope}} = -0.0188$ makes the scale grow by about 1.9 per cent per five-year period, roughly a third over the life span, so the technology partially offsets the steeply rising depreciation schedule of the first step rather than compounding it: care is estimated to be somewhat more restorative at the ages where there is more to restore. The flow value of being alive is $\hat{\chi} = 29.2$, and since the amenity scale sits at its floor, this value together with the survival and productivity returns priced in the medical-spending condition (56) carries the entire demand for medical care. The shock size $\hat{\varsigma} = 0.283$ removes more than a quarter of the health range when a shock lands, though it is identified conditional on an arrival rate that sits at its floor and inherits that caveat. The health-productivity slope is interior at $\hat{\varrho} = 0.983$, high in its unit interval: the earnings channel is the dominant use of health in the fitted model.

Two parameters at the boundary. The two boundary values are individually uninformative as point estimates, but jointly they are coherent, and the coherence is the

finding: the estimator switches off both of the channels through which health matters other than earnings and survival. The amenity scale sits at its lower bound, which switches the direct utility value of health off. That corner is not an artifact of where the bound was placed—relaxing it by two orders of magnitude sends the value a further twenty-four-fold down while the objective stays essentially flat in that direction. The shock arrival rate sits at its floor, switching health risk off, and we read this as censoring rather than as evidence that health risk is absent: the dispersion and persistence facts that motivate the shock are second moments of spending, which the target set excludes for the measurement reasons given above, and the case for the stochastic specification rests on the evidence of Section 5.2 rather than on the fitted arrival rate. Both corners are unchanged when the education channel in survival is respecified, so they are properties of the health block against these data rather than of any single estimation pass. In the fitted model, health matters through what it earns and through how long it lets one live, not through how it feels and not through the risk of losing it.

Table 5: Moment fit at $\hat{\Theta}_{MD}$ (general-equilibrium prices).

MOMENT	DATA	MODEL	MODEL/DATA
Labour			
Prime-age male hours (convention)	0.333	0.326	0.98
Medical spending			
Income elasticity of m , pooled	0.575	0.724	1.26
Income elasticity of m , within age	0.573	0.831	1.45
Spending share, ages 25–34	0.0182	0.0211	1.16
Spending share, ages 50–64	0.0231	0.0225	0.97
Spending share, ages 65–79	0.0332	0.0203	0.61
Log-slope of mean m , ages 25–79	0.0675	0.0885	1.31
Earnings			
Skill premium, ages 25–34	0.659	0.717	1.09
Survival			
Five-year survival, low education, 50–64	0.951	0.995	1.05
Five-year survival, high education, 50–64	0.972	0.998	1.03

Notes. Model moments are computed from the stationary distribution at $\hat{\Theta}_{MD}$, evaluated at the baseline’s own general-equilibrium prices and dependency-matched population growth—the configuration the experiments run in. The education–survival *difference* implied by the last two rows is right-signed and about one seventh of the measured gap; Section 4.6 reports it.

Reading the fit. Half of the moments land within ten per cent of their targets: hours (0.98), the 50–64 spending share (0.97), the young skill premium (1.09), and the two survival levels in proportional terms (1.05, 1.03). The misses concentrate in the spending block and run in one direction. Both income elasticities overshoot (1.26 pooled, 1.45 within age), the age slope of spending overshoots (1.31), and the age profile of the

spending shares is tilted the wrong way: the young share overshoots by 16% while the old-age share undershoots by nearly two fifths (0.61). Under the proportional objective these terms dominate the residual—the within-age elasticity and the old-age share together account for nearly two thirds of it, while the two survival levels carry under one per cent.

The pattern has one economic reading, and it is the constraint under which the rest of the paper’s health results should be read: at the parameter values that fit the age profile of medical-spending budget shares, the model’s income-responsiveness of medical spending is too strong, and the age profile of its spending *share* is too flat. Spending in the fitted model rises with age because depreciation rises with age, and it rises somewhat more steeply than measured spending does; but consumption rises faster still, so the share fails to tilt upward as it does in the data.

4.6 Validation

This subsection reports what the calibrated model delivers on objects it was not asked to match, and one object it was built to carry. Every comparison is between the model and something observable—the price of survival, counted deaths, earnings—and never between the model’s latent health index and a survey health measure, whose scales share no mapping. Table 6 collects them.

Table 6: The calibrated model against observables.

	MODEL	DATA
Panel A. Untargeted		
Value of a statistical life, multiple of output per capita	179	152
Education gap in five-year survival (H–L), ages 50–64	+0.003	+0.021
Panel B. Carried from the first step		
M–F log earnings gap, ages 25–34	+0.393	+0.290
M–F log earnings gap, ages 35–49	+0.420	+0.382
M–F log earnings gap, ages 50–64	+0.370	+0.362

Notes. Panel A: neither row is a target. The value of a statistical life enters no moment. The survival *gap* is not in the objective, though the two survival *levels* whose difference it is are (Table 3); proportional weighting on levels near one makes the sign of their difference nearly costless (Appendix B). The benefit-transfer reference is [Viscusi and Masterman \(2017\)](#); the survival gap is from ENASEM. Panel B: the earnings gaps are not untargeted tests—gender enters the model through the measured first-step profiles, which are constructed so that the model reproduces the measured gaps—and are reported to show that the construction works; data from ENIGH 2016–2018, earners, expansion-weighted. Model values at the baseline’s own general-equilibrium prices.

The price of survival. The felicity carries a flow value of being alive because a bare GHH–CRRA specification cannot price survival: its normalization places the flow

utility of a living household at the same zero assigned to death, which mechanically delivers a near-zero willingness to pay for survival that no amenity scale repairs. That parameter is estimated from the spending moments it shifts, as Section 4.4 describes, and not from any target on the value of life; the natural such target is a benefit transfer whose standard error is a judgment band rather than a sampling error, and so soft a target would function as a normalization rather than a restriction.

The value of a statistical life is therefore an output. We compute it from the value function as the mass-weighted trimmed mean, over the middle fifth of the population distribution, of the marginal rate of substitution between mortality risk and money, expressed as a multiple of annual output per capita, which is the scale-free form in which the benefit-transfer literature makes the number comparable across countries. At the estimates the model delivers 179 times output per capita against an external reference of 152 (Viscusi and Masterman, 2017), about a sixth above. Given the estimation penalty that rules out negative valuations of survival, no household carries one; the level of the valuation itself is unpenalised and untargeted. It is the sharpest such check the calibration passes: a preference parameter disciplined only indirectly, through the medical spending it induces, lands about a sixth above an external valuation of life that entered no moment. The welfare statements of Section 6 rest on consumption and the labour wedge rather than on the monetized value of survival, and do not depend on this number.

The education gap in survival. The model’s education gap in five-year survival at ages 50–64 is right-signed and small: +0.30 percentage points against +2.09 in the data, about one seventh of the measured magnitude. The decomposition required by the second interpretation rule of Section 4.4—computed by holding policies and the population distribution fixed and substituting the pooled survival schedule—attributes +0.24 percentage points of the +0.30 to the measured schedules and +0.06 to the endogenous investment channel. The sign is therefore earned by measurement rather than generated by the mechanism, and the shortfall in magnitude has a located cause: the education gradient of the first-step input is clipped by the cap discussed in Section 4.3 at exactly these ages, and the model’s near-ceiling health leaves no endogenous margin to replace what the cap removes. What the model does deliver is the integrated ordering—expected age at death higher for the high-education group by 2.5 years for men and 1.9 for women—and it is that ordering, not any mortality level, on which the pension-incidence results of Section 6 rest. Section 5.2 takes the shortfall up as the health block’s binding specification question.

The gender structure of earnings. The three male–female earnings gaps in Panel B are not untargeted tests, and reporting them as such would misstate what the model does. Gender enters the model only through measured first-step inputs—the sex-specific efficiency profiles and survival schedules and the sex-by-skill birth shares—and no

structural parameter differs by sex. The first step divides the estimated log-earnings objects by $1 + 1/\nu_\ell = 1.5$ precisely so that the model reproduces the measured earnings gaps rather than amplifying them through GHH labour supply a second time. The model is therefore disciplined to carry the gender structure of earnings; the rows show that the construction works. It reproduces the age hump—the gap rising into 35–49 and easing at 50–64—with levels overshooting throughout and most in the youngest band, an overshoot that most plausibly reflects earner selection embedded in the female profile input rather than anything in the model. This is what the incidence results of Section 6 are built on, so it matters that the gaps are carried rather than assumed.

Second moments of spending. One class of observable is outside the model’s reach and is neither targeted nor reported against it. Health production satisfies an Inada condition, so no household in the model ever chooses zero medical spending, whereas most of the survey’s households report none over its three-month recall window; and the model’s period is five years, so “spent nothing in three months” and “spent nothing in five years” are different events. The cross-sectional dispersion and the persistence of measured spending are accordingly comparisons of the model’s signal against the data’s signal-plus-noise rather than tests of fit. Bringing them into scope requires a transitory-expenditure or measurement-error layer, which we leave to future work.

4.7 The Calibrated Baseline Equilibrium

Solving the full stationary general equilibrium at the estimates yields the calibrated Mexican economy of Table 7. This is the equilibrium whose prices every number in Sections 4.5–4.6 is evaluated at, and it is the paper’s current baseline.

Table 7: Baseline general equilibrium at $\hat{\Theta}_{MD}$ (México, 2020): stationary aggregates.

QUANTITY	VALUE
Stationary aggregates and prices	
K (capital, per cohort mass)	5.16
L (effective labour)	6.51
Y (output)	9.49
r (annual)	8.58%
w (wage)	0.875
τ^p (PAYG contribution rate)	5.66%
$\overline{pen}$ (pension flow per retiree)	0.475
C/Y	61.1%
M/Y	3.21%
N^R/N^W (PAYG dependency)	0.141
$5K/Y$ (annual capital–output ratio)	2.72
$5B/Y$ (annual debt–output ratio)	112%

Notes. Stationary equilibrium at the estimates of Table 4, on the estimation grid. τ^p clears the PAYG block and B is the residual that closes the general budget, so neither is a free parameter. Lifetime utility at birth is not tabulated: its level is not interpretable, and only its *change* under a counterfactual carries welfare content (Section 6.2).

Four of these deserve comment. The old-age dependency ratio comes out at 0.1414, equal to the 2020 census value—but this is matched by construction, not validated: the population growth rate is re-bisected to the dependency target whenever the survival schedules change (Section 4.3), so the row records a discipline of the calibration, not evidence for the model. Three of them are misses worth naming. The annual capital–output ratio is 2.72 against a Mexican value near 3.29, a 17 per cent shortfall; the investment rate overshoots, at 22.6 per cent of output against 20.2 in the data; and the implied public debt, at 112 per cent of annual output against a Mexican federal debt near 50%, is the residual of the closure, since B absorbs whatever stationary stock balances the primary account. The investment overshoot moves with the high population growth rate and the long lives this calibration carries. The third is a property of the closure rather than a fitted object, and the natural refinement is to target a debt ratio and let a tax instrument clear the budget instead.

This is the economy the reader should carry into Section 6: a calibrated México in which buffer-stock saving is weak because idiosyncratic income mean-reverts, the contribution rate clears the pension system at 5.66%, and the four types are separated by the ENIGH skill premium of the first step (Table 2). The experiments of Section 6 perturb exactly this object.

5 Medical Spending and the Health Block

This section describes what Mexican households buy when they buy medical care, and then reports what the estimated health block of the model delivers against that description and where it falls short.

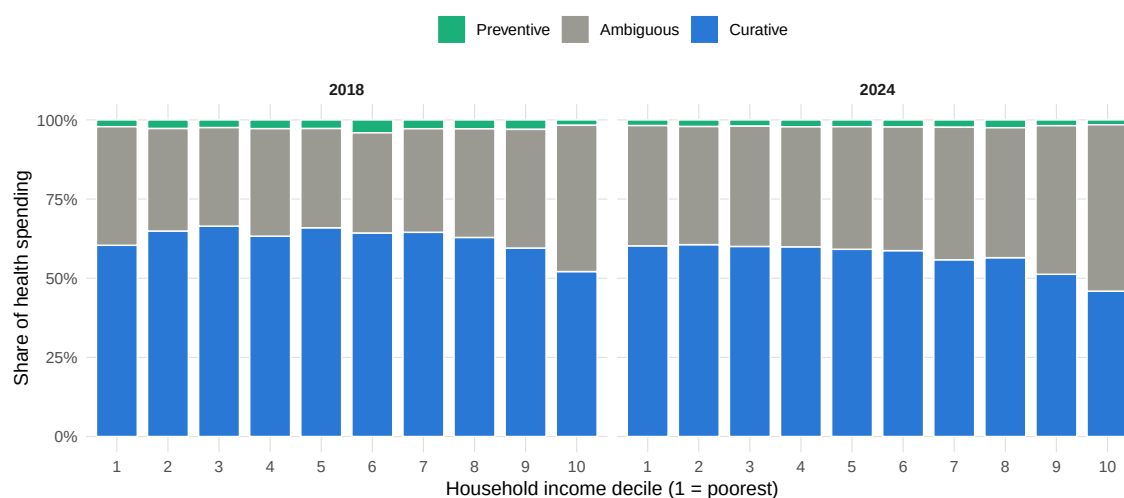
5.1 What Mexican Households Buy When They Buy Medical Care

One convention governs the empirical evidence that follows. ENIGH records out-of-pocket medical expenditures but captures only between 22 and 34 percent of the corresponding total in the national health accounts. We therefore use ENIGH to characterize the shape, gradients, and composition of medical spending, but not its level. In particular, we do not interpret any peso amount in ENIGH as a national aggregate, and none of our results depends on the absolute magnitude of medical spending. The same principle applies to the spending moments used in the calibration in Section 4, which are expressed as shares, slopes, and ratios.

The first fact concerns composition. We classify each out-of-pocket expenditure item, using an auditable rule, as unambiguously preventive, unambiguously curative, or ambiguous. This procedure yields a transparent bracket rather than a point estimate: in 2024, 2.0 per cent of spending is unambiguously preventive, 53.6 per cent is unambiguously curative, and the remaining 44.5 per cent—over-the-counter medicines, dental and optical care, and other items whose intent cannot be inferred from the expenditure classification—is reported as ambiguous rather than allocated by assumption ((Figure 1). This bracket supports a single robust conclusion: Mexican out-of-pocket medical spending is majority curative under any assignment of the ambiguous category. The classification does not identify how much of the ambiguous spending is preventive, nor does it need to for this conclusion, since even assigning the entire ambiguous category to prevention leaves curative spending as the majority.

Observed OOP health spending is dominated by curative items in every decile

Composition of household OOP health spending by income decile (auditable rubro classification)



Source: ENIGH 2018 (J-rubros) and 2024 (COICOP rubros), classification in output/tables/curative_preventive_classification.csv. Ambiguous (OTC medicines, dental, optical, chronic maintenance, diagnostics, insurance premiums) is kept as its own category; assigning it wholly to curative or wholly to preventive brackets the preventive share between ~2% and ~46% (see sensitivity table). 2024 rubros do not record prescription status, which shifts items from ambiguous to curative relative to 2018.

Figure 1: Composition of out-of-pocket health spending by income decile, ENIGH 2018 and 2024. Every expenditure item is assigned to one of three classes by an auditable rule; the ambiguous block (over-the-counter medicines, dental, optical, chronic-disease maintenance, unqualified diagnostics, insurance premiums) is reported rather than allocated. Assigning it entirely to prevention leaves curative spending the majority; assigning it to treatment makes 2024 spending 98.1 percent curative. The two waves are classified on different schemes—2018 on the ENIGH J-rubros, 2024 on COICOP, which does not record prescription status—so the composition is comparable across deciles within a wave but not across waves. Source: ENIGH 2018 and 2024 (INEGI), authors’ calculations.

The second fact is that how much of that care a household buys tracks what it can pay. Among working-age households with labour income, the pooled income elasticity of out-of-pocket medical spending is 0.575 (Table 3): ten per cent more labour income is associated with under six per cent more spending. Two properties of that number matter, and they are different properties. It is positive, so spending tracks resources and not need alone: were care bought purely as illness dictated, means would not predict how much of it a household ends up paying for. And it lies below one, so medical outlays absorb a larger share of income the poorer the household.

Figure 1 adds a second income gradient, in composition rather than level. In 2024 the curative share runs near 60 per cent across the bottom half of the distribution and falls to 46 per cent in the top decile, while the ambiguous block rises from about 38 to about 53 per cent; moving up the distribution adds disproportionately the care whose purpose the receipt cannot resolve. That is a limit on what the classification can settle, not evidence against the majority it establishes.

Two qualifications belong with that number. It is a household cross-section of out-of-pocket payment under public coverage, not the aggregate development elasticity of Hall and Jones (2007), and the two are not in competition. And it is a moment the estimation targets: Section 4.5 reports that the fitted model overshoots it by about a

quarter. What this paragraph uses is the measurement, not the fit.

The implication for what follows is structural rather than decorative: if measured spending is predominantly a response to realized illness, and how much of that response a household buys tracks the resources it has, then a specification in which restoration is unconstrained—available to every household on the same terms whenever a draw lands—is a strong assumption rather than a neutral one.

5.2 What the Health Block Delivers, and What It Does Not

This evidence bears directly on the specification in Section 3.4.3, which departs from the constant-returns deterministic rule along two dimensions: health repair is scaled by the remaining health deficit, and health is subject to a shock. Both departures address a quantitative failure of the constant-returns specification. Given the depreciation schedule estimated in Section 4.3, that specification produces an essentially degenerate working-age health distribution: more than 99 per cent of the working-age population is concentrated at the health ceiling. With five-year depreciation of only one tenth of one per cent, virtually any household with positive medical spending is restored to full health. This creates two problems. First, the mechanism through which income inequality generates health inequality has virtually no variation on which to operate, because within-cohort health dispersion is zero over most of the life cycle. Second, medical spending collapses: with almost no health deficit to repair, the spending share among young households falls to roughly three per cent of its empirical counterpart.

Neither departure is sufficient on its own. Deficit-scaled repair makes the marginal product of medical care decline with health, restoring an interior motive for spending and raising medical expenditures toward their empirical level. Without health risk, however, households within a cohort evolve similarly and the cross-sectional health distribution remains compressed. Introducing the health shock generates within-cohort dispersion, but under constant-returns repair this dispersion is short-lived: households hit by an adverse shock can spend their way directly back to full health. The two departures are therefore complementary. Together, they reduce the share of the working-age population at the health ceiling from more than 99 per cent to about 75 per cent and bring the spending share of young households to within a few per cent of its empirical target. This is the specification subsequently disciplined by the estimation in Section 4.4.

The implications of the estimation for the education gradient in survival are reported in Section 4.6. The model generates a survival gap with the correct sign, but its magnitude is only about one seventh of the observed gap, and most of this difference is inherited from the measured survival schedules rather than generated endogenously through health investment. The source of this shortfall is apparent in the health stock. Across all four types and all seventeen age bands, health lies in the narrow interval $[0.990, 1.000]$: households enter at the ceiling and remain within one per cent of it

throughout the life cycle. The ordering within this interval is as expected—at ages 60–64, health is 0.9970 for high-skill men and 0.9969 for high-skill women, compared with 0.9963 for low-skill men and 0.9949 for low-skill women—but these differences are too small to generate substantial mortality differences, regardless of the estimated survival return to health. In this configuration, medical spending primarily serves to maintain health rather than to repair substantial health deficits. Spending rises with age because depreciation increases with age and households must devote more resources to remaining near the health ceiling. Its skill gradient therefore largely reflects an income gradient in the resources devoted to maintaining health, rather than meaningful differences in the health stock attained.

The concentration of health at the ceiling is a property of the estimated block rather than an artifact of a particular parameter vector. Varying the age slope of the health technology over its admissible range does not eliminate it: as repair becomes more costly with age, households respond by increasing medical spending several-fold to remain near the ceiling rather than allowing health to deteriorate, and the resulting spending profiles exceed those in the data by an order of magnitude. The same limitation arises with an additive education shifter in the survival modulation, which an earlier specification included as a tenth parameter but which we ultimately replaced with the measured schedules. Because the modulation cap binds for roughly ninety-nine per cent of the high-education mass at the relevant ages, the survival gap generated by such a shifter saturates at about four per cent of the gap in the data, while low-education survival remains near 0.996, compared with 0.951 in the data. Matching the observed survival level would therefore require a substantial share of households at ages 50–64 to lie well below the health ceiling. Achieving that dispersion requires a change in the model specification rather than different moments or parameter bounds.

The estimated block therefore supports a deliberately bounded claim. At parameter values that jointly match the spending profile, both income elasticities, the skill premium, and survival levels, the health-investment mechanism generates almost none of the observed education–survival gradient. Working-age depreciation is too small for differences in health investment to produce meaningful health disparities before mortality becomes quantitatively important, while the survival modulation largely eliminates the longevity return to additional investment for households already near full health. Accordingly, no counterfactual in this paper attributes the education–survival gap to the investment channel, and the fiscal results in Section 6 do not operate through that mechanism. What the estimated block does deliver is the mortality ordering required for the incidence analysis—expected age at death is higher for the high-education group by 2.5 years for men and 1.9 years for women—together with the earnings return to health identified by the skill premium.

One interpretation of this shortfall is particularly informative because it points to the model specification rather than to the estimated parameters. In the model, medical spending is chosen before the health shock is realized and is therefore preventive by

construction: households invest in health before illness occurs. By contrast, the evidence above indicates that Mexican out-of-pocket medical spending is predominantly curative, reflecting expenditures incurred after illness has arrived. These two forms of spending differ in their timing, their exposure to expenditure risk, and the way in which spending gradients translate into survival gradients. Preventive investment accumulates through the health stock over many periods, whereas curative spending operates only after adverse health shocks occur and only for the households that experience them. This distinction suggests two natural directions for extending the model: shifting the health-investment mechanism toward older ages, where estimated depreciation is substantially larger, or modifying the repair technology so that mid-life health remains meaningfully below the ceiling. Which of these mechanisms better captures the data is the central specification question for the next stage of the project.

6 Experiments

México’s old-age dependency ratio is projected to more than double between 2020 and 2050, from 0.141 to 0.307 (CONAPO, 2023). In the model this projection has an immediate fiscal translation, because the PAYG identity (23) ties the contribution rate to the dependency ratio one for one: at the baseline replacement rate $\kappa = 0.40$, a dependency ratio of 0.307 means a contribution rate of 12.3%, against 5.66% today. The question this section asks is what that arithmetic costs, who pays it, and what the two classical pension margins—cutting benefits and raising the retirement age—buy relative to doing nothing.

We answer it with three general-equilibrium experiments on the calibrated baseline of Section 4.7. All three impose the same 2050 demographics. The first, *inaction* (CF0), changes no policy and lets the contribution rate absorb the entire demographic shift. The second, the *benefit margin* (CF1), cuts the replacement rate until the contribution rate returns to its 2020 value. The third, the *retirement margin* (CF2), raises the retirement age from 65 to 70 and leaves benefits untouched. The PAYG identity organizes the whole section: since $\tau^p = \kappa N^R/N^W$, a government facing a doubling dependency ratio must give up one of three things—the contribution rate, the replacement rate, or the boundary between N^W and N^R —and the three experiments move exactly one each. Inaction is not the absence of a policy; it is the choice of the contribution rate as the adjusting margin, and we treat it as the tax arm of the menu.

Every experiment is a stationary equilibrium computed on the baseline model without modification: the estimated parameter vector $\hat{\Theta}_{MD}$ and every externally sourced parameter are held fixed, and the contribution rate, factor prices, the pension benefit, and the stationary debt stock all clear endogenously in each economy. At the baseline inputs the counterfactual driver reproduces the baseline equilibrium of Table 7 exactly, field for field, so every difference reported below is attributable to the imposed

change alone. All comparisons are between stationary equilibria—no transition path is computed—and all equilibria are solved on the estimation grid of Appendix B; both qualifications matter for how the results may be read, and we return to them where they bind.

6.1 Design

The aged economy. Aging enters through the population growth rate alone. The baseline disciplines n_p so that the model’s stationary age distribution reproduces the 2020 census dependency ratio (Section 4.3); the aged economy re-solves the same bisection against CONAPO’s projected 2050 ratio of 0.3065, which lowers n_p from 3.51% to 1.48% per year. The counterfactual n_p is therefore not a fertility forecast: it is the growth rate at which the model’s stationary demography matches the projected 2050 population structure, exactly as the baseline n_p matches the 2020 structure, and both ends of the comparison are disciplined by the same source (Appendix C).

We do not impose projected 2050 survival schedules, for a reason specific to this model: survival is partly endogenous. The measured schedules $\psi_j^{\circ, g, \theta}$ are modulated by the health stock through (11), and the health stock responds to prices and income. A “2050 survival schedule” imposed from outside would therefore bake the behavioral responses of some other equilibrium into what is meant to be an exogenous environment change, and the counterfactual would no longer be clean. Fertility decline is in any case the dominant driver of México’s projected aging, so the fertility channel carries the experiment; the survival schedules, and the birth composition $\pi_1^{g, \theta}$, stay at their 2020 values throughout, and we disclose both choices wherever the results depend on them.

The two reforms. Both reforms share the aged economy’s demographics, so each is read against inaction, not against 2020. In CF1 the replacement rate is set analytically from the PAYG identity—the value that returns the equilibrium contribution rate to its baseline 5.658%—and confirmed in equilibrium; the resulting $\hat{\kappa}$ is itself a result—it measures how much benefit must be given up to hold the contribution rate fixed against a doubling dependency ratio. In CF2 the retirement period moves from $j_R = 10$ to $j_R = 11$, so households work through ages 65–69, and κ stays at 0.40; the contribution rate is left free, and where it lands measures how much of the fiscal cost of aging longer working lives recover. One input must be added for CF2: the age-efficiency profiles of Section 4.3 end at ages 60–64, because the ENIGH estimation sample ends at the baseline retirement age. We extend each of the four profiles to ages 65–69 with the same regression re-estimated on ENIGH earners aged 20–69, taking only the within-regression 65–69 to 60–64 growth ratio per type, so that the staged profiles at all other ages are untouched. The estimated ratios lie between 0.87 and 0.96—late-life efficiency declines 4–13% from its 60–64 level, continuing the measured trend—but they come

from the roughly one in five persons of those ages who still work (19.2% employment at 65–69, against 28.5% at 60–64), a selected group. The extension is therefore an upper bound on the productivity of the marginal worker the reform forces into employment, a bias we return to when reading CF2’s incidence.

Why these three, and not others. The menu is short because the identity that generates it is short: $\tau^p = \kappa N^R/N^W$ has three arguments and no others, so within the pension system the three experiments are a closed taxonomy of responses, not a sample of interesting policies. What lies outside it does so for stated reasons. Financing the pension gap from general revenue moves no argument of the identity; it relocates the obligation to a different budget constraint and raises a fiscal-sustainability question the paper’s closure does not address (Section 6.8). Restructuring the benefit rule—linking benefits to individual contribution histories, or changing indexation—changes the object rather than the margin: the model’s benefit is flat and wage-indexed, as Section 3 discloses, and a contributory link would be a different model. Coverage expansion through formalization moves the contribution base itself and is designed and left as an extension; models that carry that margin, such as Joubert (2015), price exactly the response this paper’s closure holds fixed. Combinations are omitted for a different reason: their fiscal arithmetic is additive and can be read off the identity, but their incidence is not—each reform changes the prices through which the others operate—and studying the margins one at a time is what makes each one’s incidence attributable.

What the household sees. One structural fact disciplines the interpretation of the first two experiments: neither n_p nor κ appears anywhere in the household problem of Section 3.7. The household sees its environment only through the price-and-fiscal bundle $(r, w, \tau^p, \overline{pen})$. Holding that bundle at baseline values, the direct effect of aging and of the benefit cut on any household is identically zero; the entire incidence of CF0 and CF1 is general-equilibrium, transmitted through factor prices and the pension system. Only the retirement-age reform reaches the household problem directly, by changing the ages at which it can earn. Aging, in this economy, hurts nobody except through prices and the fiscal system—which is precisely why the experiments must be solved in general equilibrium rather than priced at baseline.

Before any experiment was solved, the PAYG identity fixed three predictions: $\tau^p(\text{CF0}) \approx 0.40 \times 0.3065 = 12.3\%$, $\widehat{\kappa}(\text{CF1}) \approx \tau_{2020}^p/0.3065 \approx 0.185$, and, with band 10 moved from retirees to workers, $\tau^p(\text{CF2}) \approx 8.8\%$. The equilibria confirm all three to the reported digits. The identity, however, is silent on prices, quantities, and welfare—everything in the next four subsections is what the general equilibrium adds to the arithmetic.

6.2 The Welfare Metric

Every welfare statement in this section answers one question: how much consumption would a newborn give up to be born into one economy rather than the other? The answer, λ , is the constant share of consumption—surrendered at every age and in every state of life—that leaves the better economy’s newborn exactly as well off as the other’s; a λ of five per cent means giving up one twentieth of consumption at every age, for life. It is a lifetime object, not a period one, computed along the newborn’s own equilibrium path rather than a hypothetical one. Formally, welfare comparisons across stationary economies are stated as consumption equivalents in the compensating direction of [De Nardi et al. \(2017\)](#). For two economies A and B and a newborn of type (g, θ) whose expected lifetime utility is higher in A , we report the constant fraction $\lambda_{g,\theta}$ of consumption that the newborn of the *better* economy would give up, at every age and state along its own equilibrium allocation, to be exactly as well off as the newborn of the other economy:

$$\mathbb{E}_1 \left[\sum_{j=1}^J \beta^{j-1} \left(\prod_{i=2}^j \psi_i^{g,\theta}(h'_{i-1}) \right) u((1 - \lambda_{g,\theta}) c_j^A, \ell_j^A, h_j^A) \right] = \mathcal{W}_1^B(g, \theta), \quad (51)$$

where (c^A, ℓ^A, h^A) is the equilibrium allocation of economy A held fixed. The direction is not a stylistic choice. With $\gamma > 1$ the felicity (5) is bounded above in consumption—as c grows the bracket converges to $\chi + 1/(\gamma - 1)$ —but unbounded below as consumption falls toward the labor-disutility floor, so a proportional *reduction* on the better side always reaches the target value, while the familiar increment on the worse side may not: there are gaps no amount of extra consumption can close, because part of what the better economy delivers is survival, valued through χ , which extra consumption cannot buy. Equation (51) is finite by construction and counts the survival value in full. We report λ per type and aggregated with the birth shares $\pi_1^{g,\theta}$, signed so that $\lambda > 0$ means the first-named economy is the better one.

The same object is computed by age. For each five-year band j , λ_j scales the remaining-lifetime consumption of economy A ’s age- j residents, averaged over that economy’s own stationary distribution at age j , until their remaining-lifetime value equals that of the other economy’s same-age residents. Two rules govern how these profiles may be read, and we apply both throughout. First, λ_j compares *residents of different steady states* at the same age; it is not the transition incidence on any cohort alive at a reform date, because no transition is computed. In plain terms, these profiles price destinations, not journeys: no person alive at a reform date experiences the numbers reported here. Second, at old ages consumption is a weak welfare instrument: over a short remaining horizon a large share of remaining value is the flow value of being alive, which consumption scaling does not touch, so closing a given value gap requires a mechanically large λ_j . Put plainly: near the end of life much of what remains valuable

is simply being alive, which consumption cannot buy, so closing even a small value gap takes a large consumption adjustment. The *age gradient* of the profile is the reportable object; the old-age *levels* carry this amplification and we do not lean on them.

One principle organizes every welfare number that follows. A pay-as-you-go pillar pays an implicit return equal to the growth rate of the contribution base, while private saving pays the market return (Aaron, 1966). Wherever the market return exceeds the economy's growth rate—and it does, in every economy on this menu—mandatory participation in the unfunded pillar transfers resources away from the participant; the size of the gap between the two returns is what every welfare comparison below ultimately measures. Reforms that shrink the mandatory pillar move newborns toward the higher return; forces that move factor prices move the gap itself.

6.3 The Cost of Inaction

Table 8 reports the three counterfactual equilibria against the baseline; this subsection reads the second column.

Table 8: The counterfactual equilibria: 2020 baseline, inaction, and the two reforms at 2050 demographics.

	BASILINE 2020	CF0 INACTION	CF1 BENEFIT CUT	CF2 RETIRE AT 70
Imposed				
n_p (annual, %)	3.51	1.48	1.48	1.48
κ (replacement rate)	0.40	0.40	0.185	0.40
Retirement age	65	65	65	70
Pension block (endogenous)				
N^R/N^W	0.1414	0.3064	0.3065	0.2195
τ^p (%)	5.66	12.26	5.66	8.78
$\overline{pen}$	0.475	0.569	0.303	0.580
Prices and aggregates (endogenous)				
r (annual, %)	8.58	6.82	5.89	6.82
w	0.875	0.984	1.053	0.984
K	5.16	10.04	12.82	10.96
L	6.51	9.44	10.18	10.31
Y	9.49	15.48	17.87	16.91
K/L	0.79	1.06	1.26	1.06
$5K/Y$	2.72	3.24	3.59	3.24
C/Y (%)	61.1	63.1	61.0	63.1
M/Y (%)	3.21	3.16	2.92	3.04
$5B/Y$ (%)	112	107	124	107

Notes. Each column is a converged stationary general equilibrium at $\hat{\Theta}_{MD}$ under the baseline closure (τ^p clears the PAYG block, B residually closes the general budget), on the estimation grid; levels carry that caveat, and cross-column comparisons are the reportable objects. r is the annualized pre-tax return; τ^p the five-year-period contribution rate; $5K/Y$ and $5B/Y$ annual ratios. The imposed rows are exhaustive: everything below them adjusts. In CF1, $\kappa = 0.1846$ is set analytically from the PAYG identity and confirmed in equilibrium: τ^p returns to its baseline value in a single solve (realized 5.660% against the target 5.658%). The difference between the CF0 and CF1 dependency ratios—about 1×10^{-4} —is the endogenous-survival footprint of the benefit cut. $5B/Y$ is an outcome of the residual debt closure described at Table 7; it is reported in every column and targeted in none.

Aggregates. Under inaction the contribution rate more than doubles, from 5.66% to 12.26%, and the equilibrium confirms the PAYG identity to the last reported digit: $12.257\% = 0.40 \times 0.3064$. But the same demographic force that doubles the payroll tax also deepens capital. Slower population growth tilts the stationary age distribution toward older, asset-rich cohorts, so the capital–labor ratio rises from 0.79 to 1.06; the marginal product of capital falls—the annual return drops from 8.58% to 6.82%—and the wage rises 12.4%. The two movements nearly cancel in the worker’s budget: the net-of-tax wage $w(1 - \tau^w - \tau^p)$ rises about 3.6% despite the doubled contribution rate,

and because the flat benefit (22) is indexed to the wage, the pension rises almost 20%. Aging under inaction, in general equilibrium, is not a story of impoverished workers financing unaffordable pensions; it is a story of a doubled tax wedge financed, and slightly more than financed, by capital deepening.

Newborn welfare: almost exactly nothing. The welfare consequence for a newborn is the sharpest statement of that offset. Table 9 reports the consumption equivalents: a newborn drawn from the birth distribution of the aged economy would give up 0.145% of lifetime consumption before being as badly off as a baseline newborn—an effect economically indistinguishable from zero, with mixed signs across types no larger than 0.43% in absolute value. A newborn holds no assets, so the lower return costs it little; it sells labor at a wage 12.4% higher; and it can re-optimize its entire life cycle around the new prices from the start. Decomposing through an intermediate arm that imposes the aged economy’s factor prices at the baseline fiscal bundle shows the anatomy: a substantial price-channel gain to every type, a comparable fiscal loss (the doubled τ^p net of the higher pension) to every type, and the small residual the table reports as the net; the split is ordering-dependent—prices before fiscal—though neither channel’s sign is. What would break the offset is what the model’s closure assumes: with an open-economy interest rate, or benefits not indexed to the wage, the price channel would shrink while the fiscal channel stayed, and inaction would cost newborns in full.

The economics beneath the cancellation is plain. When the population grows more slowly, labor becomes scarce relative to capital: the wage rises and the return falls. A newborn faces a doubled contribution rate but sells its labor at the higher wage for a full working life, and because the benefit is indexed to the contemporaneous wage, the pension it will eventually collect rises as well; both sides of its lifetime budget move up together and nearly cancel. The newborn is not protected by policy; it is protected by the factor-price movement that the same demographic change produces. Part of the capital deepening behind that movement is not a savings response at all. Gross investment absorbs $(n_p + \delta)K$ each period, so slower population growth frees resources that were previously equipping new workers—a widening dividend that is a consequence of the stationary-demography device of Section 6.1, and part of why the aged economy is richer per worker.

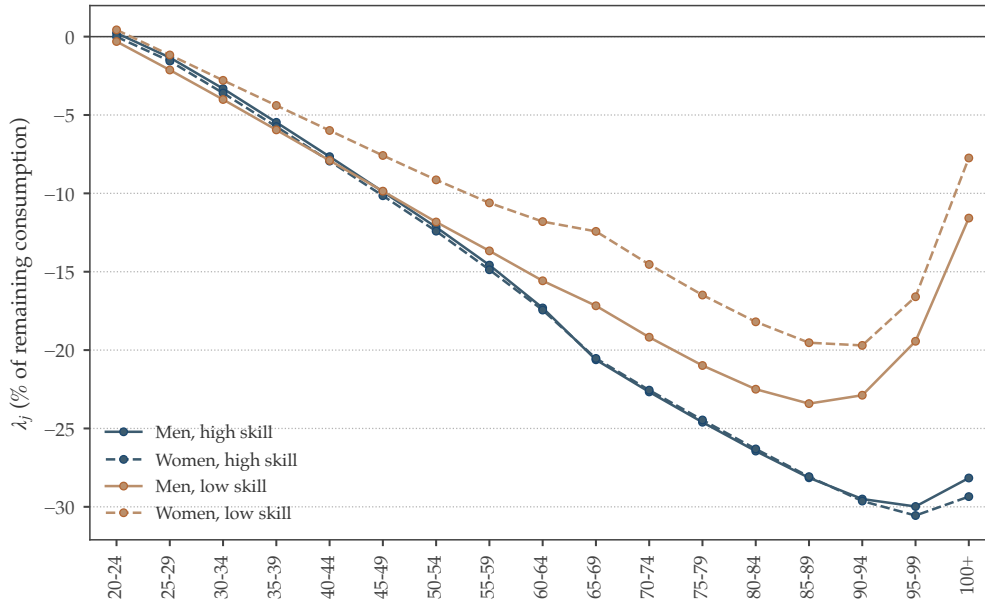
Table 9: Newborn welfare across the menu: consumption equivalents λ (% of lifetime consumption).

	CF0 vs. 2020	CF1 vs. CF0	CF2 vs. CF0	CF2 vs. CF1
Men, low skill	−0.31	+8.13	+3.00	−5.39
Men, high skill	+0.23	+9.30	+4.43	−5.28
Women, low skill	+0.43	+7.82	+2.19	−5.82
Women, high skill	−0.01	+8.58	+3.96	−4.97
Aggregate (π_1 -weighted)	+0.14	+8.29	+3.11	−5.45

Notes. Signed De Nardi-direction consumption equivalents (51): $\lambda > 0$ means the first-named economy’s newborn is better off, and $|\lambda|$ is the share of lifetime consumption the better economy’s newborn would surrender to be as well off as the other’s. Steady-state comparisons of newborns; transition generations are not modeled (see the benefit-margin discussion). Against the 2020 baseline directly, the benefit cut delivers +8.41% in the aggregate. The aggregate CF0 entry is 0.145; its numerical bracketing band is [0.1448, 0.1450].

Where the cost of inaction actually sits. Figure 2 prices the same comparison for residents of every age, and the near-zero newborn number gives way to a steep age gradient: same-age residents of the aged economy are worse off than their baseline counterparts by one to two per cent of remaining consumption at ages 25–29, six to eight per cent at 40–44, and twelve to twenty-one per cent at 65–69. The engine is the intertemporal price, not health: across all ages and types the health stock differs by less than 0.002 between the two economies—the ceiling result of Section 5.2 carries over—while the lower return rotates the entire optimal consumption profile toward youth. Residents of the aged economy consume about 7% *more* at ages 20–24 and progressively less with age—roughly 8% less in their fifties, 20% less at 65–69—with asset holdings lower commensurately, compounding at the lower return. The newborn and the age profile thus tell consistent stories: the tilt is the path a newborn *chooses* ex ante at the new prices, so it costs the newborn almost nothing, while a cross-steady-state comparison at age 70 lands on the far end of the chosen tilt and prices it as a large gap. The gradient, not the old-age level, is the finding; the levels at the oldest ages additionally carry the flow-value amplification discussed above.

Figure 2: The cost of inaction by age: consumption equivalents λ_j for same-age steady-state residents, aged economy (CF0) vs. the 2020 baseline.



Note. Signed per-age consumption equivalents: $\lambda_j < 0$ means the aged economy's age- j resident is worse off than the baseline's, by $|\lambda_j|$ per cent of remaining consumption. Skill groups: **high skill** and **low skill**; solid lines men, dashed lines women. Cross-steady-state resident comparisons, not transition incidence; old-age levels are amplified by the flow-value term (Section 6.2).

Why the incidence differs by type. The age profiles fan out in a specific order: at every age past fifty the proportional loss is largest for the two high-education types (over 20% at 65–69), intermediate for low-education men (17%), and smallest for low-education women (12%). The ordering is the return channel read through portfolio composition: inaction punishes claims on capital and rewards claims on the pension system (the wage-indexed benefit rises 20%), so a resident's exposure is the share of old-age consumption financed from private assets rather than the flat pension. High-education households, the economy's largest savers, hold the most capital-intensive old ages and bear the steepest proportional losses; low-education women, with the fewest private assets and the greatest pension reliance, are cushioned most. The flat benefit thus acts as its progressivity implies: it partially insures precisely the households least able to self-insure through wealth.

The newborn sign pattern in the first column of Table 9—low-education women the largest gainers, low-education men the only meaningful losers—traces to two measured asymmetries, and we flag its fragility explicitly. First, the pension is flat, so its 20% rise is worth most relative to lifetime resources to the lowest earners—which pushes low-education women to the top of the column. Second, the education–survival gradient enters the staged schedules with different timing by sex. The male gradient opens in the mid-fifties, whereas the female gradient is clipped to zero over ages 50–69 by the

survival-cap limitation of Section 4.3 and appears only from age 70 onward, where it is at least as large as the male one. At the estimates the model accordingly delivers educated men an expected age at death about 2.5 years above their less-educated counterparts, and educated women about 1.9 years—untargeted model outcomes, not schedule inputs. What matters for the sign pattern is the timing: because their gradient opens roughly two decades earlier, low-education men are differentiated by mortality over the whole pension-relevant range—they pay the doubled contribution for a full working life, collect the raised pension for the fewest years, and lose—whereas female mortality does not differentiate by education until age 70, so the flat benefit’s progressivity decides the female ranking alone. The sign pattern is thus the education-mortality input showing through the welfare numbers—and it is a fragile residual: each entry is a fraction of a per cent left over from price and fiscal channels an order of magnitude larger, and the deferred age-varying survival-floor refinement, which would un-clip the female 50–69 gradient, could reorder the female column. We report the pattern; we do not build on it.

6.4 The Benefit Margin

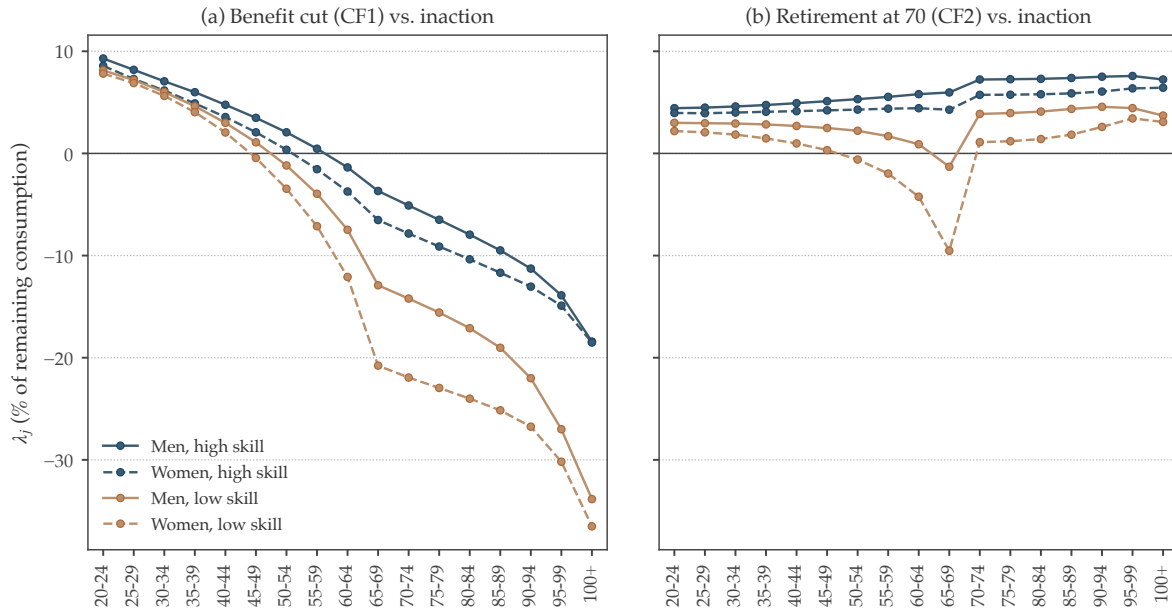
Holding the 2050 contribution rate at its 2020 value requires cutting the replacement rate from 0.40 to $\widehat{\kappa} = 0.1846$ —a 54% benefit cut, and the analytic value $\tau_{2020}^p/0.3064$ restores the contribution rate in a single step (realized 5.660% against the 5.658% target). The number itself is the first result: the identity offers no soft option, and any politically feasible partial cut buys proportionally partial relief.

Aggregates and the Aaron logic. The benefit cut is the strongest capital-deepening force on the menu. With the public pension less than half as generous ($\overline{pen}$ falls 47% relative to inaction), households self-insure old age through private saving: capital rises from 10.04 to 12.82 across the two 2050 economies, the wage gains a further 7% on top of inaction’s rise, and the annual return falls to 5.89%. For a newborn of *every* type this trade is decisively favorable: 8.3% of lifetime consumption against inaction (7.8–9.3% across types; 8.4% against the 2020 baseline). The economics is the classic Aaron condition (Aaron, 1966). A stationary PAYG pillar pays an implicit return equal to population growth—1.5% per year in the 2050 economy—while the market pays 5.9%; forcing households to route retirement claims through the low-return pillar is a standing tax on their life-cycle saving, and shrinking the pillar refunds it. General equilibrium amplifies rather than dampens the direct incentive, since the induced capital raises the wage newborns sell into. The result would flip where the condition fails—an economy near dynamic inefficiency (r near n), or one where binding borrowing constraints keep the young from doing the substitute saving; neither obtains at the estimates.

What the steady-state comparison leaves out. These gains are steady-state gains, and this experiment is where the restriction to steady states bites hardest. The 8.3% accrues to newborns *of the reformed steady state*, who never contributed at $\kappa = 0.40$. The transition generations—workers who contributed for decades at the old rate and would retire under $\kappa = 0.185$ —appear nowhere in this comparison, and their losses are exactly what a transition analysis of an unfunded system’s wind-down prices. The steady-state ranking says what the destination is worth, not what the journey costs, and the two must not be conflated: nothing in Table 9 licenses the claim that cutting benefits is a Pareto improvement.

Age incidence: a sign flip ordered by pension dependence. The closest steady-state object to that transition intuition is the age profile, panel (a) of Figure 3. Support for the benefit cut falls monotonically with age, crosses zero in mid-life, and turns severely negative for retirees: newborns and young workers gain 7–9%, and 65–69-year-olds are worse off by 3.7–21% of remaining consumption, the losses deepening through the oldest ages. A resident’s stake is the ratio of already-accumulated pension claims to the years of low- τ^p wages still ahead, so the crossing age orders by pension dependence—and the cross-type ordering confirms the mechanism exactly. Low-education women, the most pension-reliant type, flip first (by ages 45–49) and lose the most at 65–69 (–21%); low-education men flip around 50–54 and lose 13%; high-education women flip by 55–59; and high-education men, the economy’s best-buffered savers, flip last (60–64) and lose least (–3.7%). The mirror image of inaction’s incidence is exact: inaction taxes private wealth through the return, so the asset-rich lose most; the benefit cut taxes pension claims, so the pension-reliant lose most. The two policies do not differ mainly in how much welfare they deliver—they differ in *whose* old age bears the demographic bill.

Figure 3: The reform menu by age: consumption equivalents λ_j for same-age steady-state residents, each reform vs. inaction (CF0) at 2050 demographics.



Note. Signed per-age consumption equivalents vs. inaction: $\lambda_j > 0$ means the reform economy's age- j resident is better off. Panel (a): the benefit cut ($\kappa: 0.40 \rightarrow 0.1846$). Panel (b): retirement at 70. Skill groups: **high skill** and **low skill**; solid lines men, dashed lines women. Cross-steady-state resident comparisons, not transition incidence; old-age levels carry the flow-value amplification (Section 6.2). In panel (b), the 65–69 losses of the low-skill types are likely understated: the late-life efficiency input is estimated from the selected minority still working at those ages.

6.5 The Retirement Margin

Raising the retirement age to 70 moves the 65–69 band from beneficiaries to contributors, and the PAYG identity again does the fiscal arithmetic in advance: the dependency ratio falls to 0.219 and the contribution rate lands at 8.78%—almost exactly the predicted 0.40×0.219 —recovering just over half of the fiscal cost of aging while leaving the benefit rule untouched. The pension per retiree actually edges *up* relative to inaction (more contributors per beneficiary at an unchanged replacement rate on a higher wage base).

A scale expansion, not a price shock. What distinguishes this reform in general equilibrium is what does not move. Against inaction, the return is unchanged to three digits (6.819 vs. 6.820%) and the wage to within five parts in 10^5 . Extending working lives raises effective labor by 9%, and the extra labor income induces almost exactly the capital that keeps the capital–labor ratio—and hence both factor prices—where inaction put them: the economy simply scales up (Y rises from 15.48 to 16.91) and the fiscal burden falls. Where inaction and the benefit cut operate through prices, the retirement margin operates through quantities plus the budget: fiscal relief at constant

prices.

Newborn ranking. For newborns the reform is worth +3.1% of lifetime consumption relative to inaction—but −5.5% relative to the benefit cut, and the ranking $CF1 > CF2 > CF0$ holds for every one of the four types (Table 9). The reason it forfeits most of the benefit cut’s gain is the Aaron logic again: working longer *preserves* the unfunded pillar at $\kappa = 0.40$, so the newborn keeps routing a large claim through the low-return system, and pays five more years of contributions and work disutility for the privilege of the smaller wedge. The two reforms are not close substitutes for a newborn; they are a large benefit-side gain versus a moderate quantity-side gain.

Incidence: the extended work band pays. Panel (b) of Figure 3 shows an incidence pattern unlike either other experiment: not a monotone age gradient but a dip, bottoming exactly at the extended work band. Low-education women lose 9.5% of remaining consumption at 65–69 and are in losses from ages 50–54 onward—the approaching work requirement is already priced into their remaining-lifetime value—while low-education men dip to −1.3% at 65–69 only. High-education residents of *every* age gain (+3.9 to +7.6%), and retirees 70 and older all gain (a slightly higher pension, no work requirement). The heterogeneity is the late-life efficiency input read through the wage: at 65–69 the estimated efficiency of high-education men and women (1.80 and 1.39 efficiency units) still commands substantial earnings, so their extended years are well paid and the fiscal relief dominates; low-education women, at 0.64—the lowest cell in the profile—are made to work five more years at wages that barely compensate the disutility, and low-education men (0.88) sit in between. The burden of a higher retirement age falls on exactly the workers whose late-life market productivity is lowest.

Two facts qualify these losses, in opposite directions. They are steady-state numbers, so the disruption to cohorts near 65 at a reform date is not in them. And they are likely *understated*: the 65–69 efficiency input comes from the selected fifth of that age group that chooses to work, whereas the reform compels the other four fifths—including those who stopped precisely because their market productivity or health made work unattractive—so the marginal worker’s efficiency is below the estimate, and the true 65–69 losses, especially the low-education women’s −9.5%, are a lower bound in magnitude.

6.6 Behavior and Equilibrium: A Decomposition

Each experiment’s welfare effect can be split into what households’ own re-optimization delivers at unchanged aggregates and what arrives only through equilibrium. We resolve the household problem under each experiment’s direct changes while holding the entire price-and-fiscal bundle $(r, w, \tau^p, \overline{p\epsilon n})$ at baseline values, an arm in which no market clears, and then let an intermediate arm impose the experiment’s factor prices

while keeping the baseline fiscal bundle. The three differences that result attribute the total newborn-welfare change to a direct behavioral component, a factor-price channel, and a fiscal channel; the finer split prices the price channel before the fiscal one and is ordering-dependent, though no sign in Table 10 is. The decomposition is exact in units of lifetime value, in which the arms are additive; consumption equivalents are not additive across arms, so the table reports value units and the reader should take from it signs and proportions rather than magnitudes.

Table 10: Decomposition of the newborn welfare effect: behavior versus equilibrium.

	Direct (behavioral)	Factor prices	Fiscal	Total
CF0 (inaction)	0 (structural)	+3.247	−3.180	+0.067
CF1 (benefit cut)	0 (structural)	+5.017	−1.035	+3.983
CF2 (retirement age)	−0.432	+3.310	−1.348	+1.530

Notes. Birth-weighted newborn lifetime value, all arms measured against the 2020 baseline (level 340.60); rows sum by construction up to the displayed rounding. Direct arm: the experiment's non-price changes at the baseline ($r, w, \tau^p, \overline{pen}$); factor-price channel: the experiment's (r, w) at baseline fiscal variables, net of the direct arm; fiscal channel: the remainder. Budgets do not clear inside the arms. The split is ordering-dependent (prices before fiscal); no entry's sign is.

The first two rows put numbers on the structural fact of the design (Section 6.1): because neither n_p nor κ enters the household problem, the direct effect of aging and of the benefit cut is identically zero, and everything those experiments do to anyone is done by markets and the pension system. For inaction the table restates the newborn-indifference result as channel arithmetic—a price channel worth +3.25 against a fiscal channel worth −3.18, cancelling to the +0.067 residual whose smallness Section 6.3 priced at 0.145% of consumption—and it locates the benefit cut's entire +3.98 in the same two channels: the crowd-in of private saving deepens capital beyond what demography alone achieves (+5.02 through prices, against −1.03 through a 2050 fiscal bundle whose contribution rate matches the baseline by construction but whose pension is less than half). The Aaron gain of Section 6.4 arrives through the price system, not through the fiscal ledger.

The third row is the reason the decomposition earns its place. The retirement-age reform is the only experiment that reaches households directly—and its direct effect is a welfare *loss*. Compelled to work through ages 65–69 at unchanged prices and fiscal variables, the newborn is worse off by 0.43: five additional years of work disutility, sold at late-life efficiencies of 0.64–1.80, are not worth their wages when nothing else improves. Every unit of the reform's value is general-equilibrium: the scale expansion holds factor prices at inaction's values while the added labor income widens the tax base and shrinks the wedge, and the equilibrium channels (+3.31 through prices, −1.35 through the 2050 fiscal bundle) must first repay the direct loss before the reform nets its +1.53. A planner who read the retirement margin as a household-level instruction—

work longer, be better off—would have its sign wrong; the reform works only because everyone works longer at once, and the incidence of Section 6.7's third regularity follows: the band that supplies the direct loss, low-education workers in their late sixties, is the band the equilibrium gains fail to compensate.

6.7 The Policy Menu

The three experiments assemble into a menu that one identity organizes. At 2050 demographics the government chooses which variable in $\tau^p = \kappa N^R/N^W$ gives: the contribution rate (12.26%), the replacement rate ($\widehat{\kappa} = 0.185$, a 54% cut), or the retirement boundary (age 70, $\tau^p = 8.78\%$). Inaction is itself the tax arm—financing aging with a payroll-tax increase is a policy, chosen by default—and the menu prices all three.

Three regularities summarize the section. First, *the fiscal arithmetic is exact and the welfare arithmetic is not*. The PAYG identity predicted every contribution rate and the required benefit cut to within rounding before any equilibrium was solved; nothing in the identity predicted that inaction would leave newborns unharmed (+0.14%), that the benefit cut would be worth 8.3% to them, or that the retirement margin would deliver little more than a third of that—those numbers come from the general-equilibrium price responses, above all from capital deepening. Second, *for newborns the ranking is unambiguous*: the benefit cut dominates the retirement-age rise, which dominates inaction, for every type—the Aaron condition, applied to a demographic environment where the market return exceeds population growth by more than four percentage points, rewards whichever margin most shrinks the unfunded pillar. Third, *the reforms differ most in who pays*. Inaction taxes wealth-holders through the return: its losses rise with age and are steepest for the high-education savers. The benefit cut taxes pension claims: its losses concentrate on retirees, ordered by pension dependence, worst for low-education women. The retirement-age rise taxes the extended work band: its losses concentrate on low-education workers in their late sixties, again worst for low-education women, while every other group gains. A planner reading this menu is not choosing a welfare level so much as choosing an incidence—and for the two groups that recur at the bottom of every incidence table, low-education workers and pension-dependent retirees, the ranking of the two reforms is reversed.

The second and third regularities are one fact read at two ages. Every margin on the menu reprices a stock that somebody already holds and cannot rebuild. Inaction lowers the return, which reprices the remaining lifetime of anyone holding assets, so it falls hardest on older savers. The benefit cut lowers the pension, which reprices the claims of those already retired and dependent on them. The retirement-age rise extends the work band, falling on those whose late-life productivity is lowest and who must now work at ages when they would not have. Newborns rank the three margins identically because a newborn holds no stock—no assets, no accrued pension claim—and can adjust to any reform from the start; the rankings diverge with age precisely

because older residents cannot. The choice among the three is therefore a choice of whose old age pays, not a choice of how much welfare to have.

6.8 Caveats

Five caveats bound what this section claims. First, every comparison is between stationary equilibria. The transition path—and with it the losses of the generations who contributed under the old rules, the object that dominates the political economy of pension reform—is not computed, and the benefit-margin discussion states where this bites hardest. General-revenue financing of the pension gap (letting τ^w or debt absorb the shortfall rather than τ^p) would require a different budget architecture than the self-contained PAYG block of Section 3.5 and is left as a model extension. Second, all equilibria are computed on the estimation grid; levels carry that caveat, and the cross-economy comparisons the section is built on are the reportable objects. Third, the stationary debt ratio is a model-internal residual of the closure in every column (107–124% of annual output against roughly 50% for México); it is reported in every column and targeted nowhere. Fourth, the demographic experiment moves fertility only: survival schedules and the birth composition are held at 2020, for the endogeneity reason given in the design, so the aging imposed here understates any 2050 mortality improvements; and within the staged schedules the female education–survival gradient is clipped to zero over ages 50–69 (it survives at older ages), which the in-action heterogeneity discussion flags where it matters. Fifth, the slate contains no health-policy experiment: at the estimates the health stock sits against its behaviorally defended ceiling for nearly every household (Section 5.2), so a health-side instrument acts only through the withdrawal direction, and pricing that experiment fairly awaits the health-block specification decision set out in Section 7.

7 Conclusion

Demographic aging under unchanged policy redistributes rather than impoverishes. That is what this paper finds for México, in a closed economy where factor prices absorb the demographic shift. The corollary answers the second question we asked: the three margins a pay-as-you-go system can move differ far more in whose old age pays than in how much welfare the economy has. We reach both conclusions in a stochastic overlapping-generations economy estimated on Mexican micro-data, in which health is endogenous capital, four types differing by sex and education carry their own measured efficiency and survival schedules, and both factor prices and the contribution rate clear in equilibrium.

Under the 2050 demographics, imposed through fertility alone with survival held at its 2020 schedules, the equilibrium contribution rate more than doubles, from 5.66 to 12.26 per cent, and newborn welfare is essentially unchanged. Capital deepening is

why. Slower population growth raises the capital–labor ratio, so the return falls and the wage rises, and for an agent who arrives holding nothing the induced wage and pension gains nearly offset the doubled wedge. The cost falls instead on residents who already hold assets, whose remaining lifetimes are repriced at the lower return, and it rises steeply with age.

The reform menu follows from the identity that closes the system. Holding the contribution rate at its 2020 value requires cutting the replacement rate from 0.40 to 0.185, a 54 per cent cut worth 8.3 per cent of lifetime consumption to a newborn measured against inaction; raising the retirement age to seventy lands the contribution rate at 8.78 per cent and is worth 3.1 per cent on the same comparison. All four types rank the two reforms alike, and the ranking reverses with age, because each margin reprices a stock that somebody already holds and cannot rebuild. Inaction taxes wealth-holders through the return, the benefit cut taxes pension claims, and the later retirement taxes the extended work band. All of these are comparisons between stationary economies. They price destinations rather than journeys, the transition generations who contributed under the old rules are not modeled, and nothing here licenses a Pareto claim about any actual reform.

Which type pays, within each margin, is decided by heterogeneity the model measures rather than assumes. Low-education men are the only meaningful newborn losers under inaction, because their education gap in survival opens some fifteen years earlier in the staged schedules than women's, so they pay the doubled contribution across a full working life and collect the raised pension for the fewest years. Low-education women bear the higher retirement age, because their estimated late-life efficiency is the lowest of the four types and the extended work band makes them work at wages that barely compensate the effort. We flag the fragility of the first pattern where we report it. Neither statement, however, is available at all in a model without sex- and education-specific survival and efficiency schedules: quantitative models of pension reform ordinarily resolve incidence by age and by wealth, and resolving it by sex and education as well is what the apparatus is for.

For the fiscal authority this paper was written for, four statements follow and go no further. First, the menu is closed within the pension system. The identity has three arguments and no others, and everything outside them either relocates the obligation to a different budget constraint, as general-revenue financing would, or changes the structure of the benefit rule, which would be a different model. Second, the three margins differ far more in whose old age pays than in how much welfare the economy has, so the choice among them is a distributional decision presented as a fiscal one, and this paper prices the distribution. Third, a point of arithmetic rather than of result: under a flat benefit indexed to the contemporaneous wage and a fixed retirement age, longer lives raise the ratio of retirees to workers and therefore raise the contribution rate, and the identity carries no productivity term through which better population health could pay for itself. No health-policy experiment is computed here, so this is a

property of the benefit rule rather than a finding, but it does mean the fiscal case for health investment cannot be made through the pension block as currently designed. Fourth, the model's dependency ratio is demographic rather than contributor-based and the baseline is calibrated to the formal-sector population, so the fiscal pressure reported here is the favorable case. The paper ranks the margins and recommends none of them.

Four limits bound what has been shown. No transition path is computed, so no number here is incidence on any cohort alive at a reform date. The economy has neither an informality margin nor an annuity market, since decedents' assets are voided rather than bequeathed. The factor-price channel that carries the first finding is a closed-economy object, and it is the paper's most fragile assumption, because capital mobility across asynchronously aging regions would mute exactly the movement that protects the newborn (Attanasio et al., 2007). And all equilibria are computed on the same grid, so cross-economy comparisons rather than levels are the reportable objects, and the estimates are reported without standard errors, for the reasons given in Appendix B.

The agenda that follows is ordered. The binding item is the health block's specification. At the estimates the investment channel generates little of the education-survival gradient, and the endogenous health response is negligible across the compared economies, so the fiscal findings do not turn on it; whether to relocate the mechanism to old-age deterioration, where the estimated depreciation is large, or to re-specify repair so that mid-life health sits below the ceiling, is the decision that governs the next stage. After that come the transition path, which would price the journeys these steady-state comparisons cannot; a coverage-retrenchment experiment, which would put the health side of the model to work in the direction where it has bite; the formalization programme, which would open the contribution-base margin held fixed here; and the extension to Costa Rica and Panamá, whose institutional architectures differ from México's instructively.

References

- Aaron, H. J. (1966). The social insurance paradox. *Canadian Journal of Economics and Political Science*, 32(3):371–374.
- Abel, A. B., Mankiw, N. G., Summers, L. H., and Zeckhauser, R. J. (1989). Assessing dynamic efficiency: Theory and evidence. *Review of Economic Studies*, 56(1):1–19.
- Antón Sarabia, A. (2005). Average effective tax rates in Mexico. *Economía Mexicana. Nueva Época*, 14(2):185–215.
- Arrow, K. J. (1963). *Uncertainty and the Welfare Economics of Medical Care*, volume 53. Reprinted in various collections.

- Attanasio, O., Kitao, S., and Violante, G. L. (2007). Global demographic trends and social security reform. *Journal of Monetary Economics*, 54(1):144–198.
- Auclert, A., Malmberg, H., Martenet, F., and Rognlie, M. (2021). Demographics, wealth, and global imbalances in the twenty-first century. NBER Working Paper 29161, National Bureau of Economic Research.
- Auerbach, A. J. and Kotlikoff, L. J. (1987). *Dynamic Fiscal Policy*. Cambridge University Press.
- Bachas, P., Fisher-Post, M., Jensen, A., and Zucman, G. (2022). Globalization and factor income taxation. Policy Research Working Paper 9973, World Bank.
- Baeza, C. and Packard, T. (2006). *Beyond Survival: Protecting Households from Health Shocks in Latin America*. Stanford University Press, Redwood City, CA.
- Bagchi, S. (2019). Differential mortality and the progressivity of social security. *Journal of Public Economics*, 177:104044.
- Bergoeing, R., Kehoe, P. J., Kehoe, T. J., and Soto, R. (2002). A decade lost and found: Mexico and Chile in the 1980s. *Review of Economic Dynamics*, 5(1):166–205.
- Bonthuis, B. (2024). An assessment of the 2019 and 2020 pension reforms in Mexico. IMF Working Paper 2024/053, International Monetary Fund.
- Borella, M., De Nardi, M., and Yang, F. (2023). Are marriage-related taxes and social security benefits holding back female labour supply? *Review of Economic Studies*, 90(1):102–131.
- Carrasco, A. (2026). The endogenous longevity channel: Health investments, aggregate savings, and macroeconomic policy. Mimeo, Massachusetts Institute of Technology.
- Carvalho, C., Ferrero, A., and Nechio, F. (2016). Demographics and real interest rates: Inspecting the mechanism. *European Economic Review*, 88:208–226.
- CONAPO (2023). Bases de datos de la conciliación demográfica 1950 a 2019 y proyecciones de la población de México 2020 a 2070. <https://www.gob.mx/conapo>.
- Conesa, J. C. and Krueger, D. (1999). Social security reform with heterogeneous agents. *Review of Economic Dynamics*, 2(4):757–795.
- Cortés, H., Méndez, J., Molina, E., and Villarreal, H. J. (2024). Not everyone ages gracefully: Fiscal adjustments to exogenous shocks in three Latin American countries. Aix-Marseille School of Economics.
- Dalgaard, C.-J. and Strulik, H. (2014). Optimal aging and death: Understanding the Preston curve. *Journal of the European Economic Association*, 12(3):672–701.

- De Nardi, M., French, E., and Jones, J. B. (2010). Why do the elderly save? The role of medical expenses. *Journal of Political Economy*, 118(1):39–75.
- De Nardi, M., Pashchenko, S., and Porapakkarm, P. (2017). The lifetime costs of bad health. NBER Working Paper 23963, National Bureau of Economic Research.
- Fehr, H. and Kindermann, F. (2018). *Introduction to Computational Economics Using Fortran*. Oxford University Press.
- Fietz, K. et al. (2025). (In)Formalizing Jobs in Latin America and the Caribbean. World Bank, Washington, D.C.
- Fioroni, T. (2010). Optimal savings and health spending over the life cycle. *European Journal of Health Economics*, 11(3):297–312.
- Franković, I. and Wrzaczek, S. (2020). On the anatomy of medical progress within an overlapping generations economy. *Journal of Economics (Springer)*.
- Galama, T. J. and van Kippersluis, H. (2013). Health inequalities through the lens of health capital theory: Issues, solutions, and future directions. In *Research on Economic Inequality*, volume 21, pages 263–284. Emerald Group Publishing.
- García-Verdú, R. (2005). Factor shares from household survey data. Working Paper 2005-05, Banco de México.
- Gertler, P. and Gruber, J. (2002). Insuring consumption against illness. *American Economic Review*, 92(1):51–70.
- Grossman, M. (1972). On the concept of health capital and the demand for health. *Journal of Political Economy*, 80(2):223–255.
- Güvenen, F. (2009). An empirical investigation of labor income processes. *Review of Economic Dynamics*, 12(1):58–79.
- Hall, R. E. and Jones, C. I. (2007). The value of life and the rise in health spending. *Quarterly Journal of Economics*, 122(1):39–72.
- Hosseini, R., Kopecky, K. A., and Zhao, K. (2022). The evolution of health over the life cycle. *Review of Economic Dynamics*, 45:237–263.
- Hubbard, R. G., Skinner, J., and Zeldes, S. P. (1994). The importance of precautionary motives in explaining individual and aggregate saving. *Carnegie-Rochester Conference Series on Public Policy*, 40:59–125.
- Huggett, M. and Ventura, G. (1999). On the distributional effects of social security reform. *Review of Economic Dynamics*, 2(3):498–531.

- Joubert, C. (2015). Pension design with a large informal labor market: Evidence from Chile. *International Economic Review*, 56(2):673–694.
- Jung, J. and Tran, C. (2016). Market inefficiency, insurance mandate and welfare: U.S. health care reform 2010. *Review of Economic Dynamics*, 20:132–159.
- Jung, J., Tran, C., and Chambers, M. (2017). Aging and health financing in the U.S.: A general equilibrium analysis. *European Economic Review*, 100:428–462.
- Kitao, S. (2014). Sustainable social security: Four options. *Review of Economic Dynamics*, 17(4):756–779.
- Krueger, D. and Ludwig, A. (2007). On the consequences of demographic change for rates of returns to capital, and the distribution of wealth and welfare. *Journal of Monetary Economics*, 54(1):49–87.
- Kuhn, M. and Feichtinger, G. (2015). Optimal choice of health and retirement in a life-cycle model. *Journal of Economic Theory*, 158:186–212.
- Levy, S. (2008). *Good Intentions, Bad Outcomes: Social Policy, Informality, and Economic Growth in Mexico*. Brookings Institution Press, Washington, D.C.
- Lim, K. M. (2020). Public provision of health insurance and aggregate saving in an overlapping generations model with endogenous health risk: The case of South Korea. *Korean Economic Review*.
- Murphy, K. M. and Topel, R. H. (2006). The value of health and longevity. *Journal of Political Economy*, 114(5):871–904.
- Nishiyama, S. and Smetters, K. (2007). Does social security privatization produce efficiency gains? *Quarterly Journal of Economics*, 122(4):1677–1719.
- Ozkan, S. (2025). Income differences and health disparities: Roles of preventive vs. curative medicine. *Journal of Monetary Economics*, 150:103698.
- Pessino, C. and Ter-Minassian, T. (2021). *Addressing the Fiscal Costs of Population Aging in Latin America and the Caribbean, with Lessons from Advanced Countries*. Inter-American Development Bank, Washington, DC.
- Puggioni, D., Calderón, M., Cebreros Zurita, A., Fernández Bujanda, L., Inguanzo González, J. A., and Jaume, D. (2022). Inequality, income dynamics, and worker transitions: The case of Mexico. *Quantitative Economics*, 13(4):1669–1705.
- Sánchez-Romero, M., Lee, R. D., and Prskawetz, A. (2020). Redistributive effects of different pension systems when longevity varies by socioeconomic status. *The Journal of the Economics of Ageing*, 17:100259.

- Storesletten, K., Telmer, C. I., and Yaron, A. (2004). Consumption and risk sharing over the life cycle. *Journal of Monetary Economics*, 51(3):609–633.
- Summers, L. H. (1989). Some simple economics of mandated benefits. *American Economic Review*, 79(2):177–183.
- Viscusi, W. K. and Masterman, C. J. (2017). Income elasticities and global values of a statistical life. *Journal of Benefit-Cost Analysis*, 8(2):226–250.
- Yew, S. L. and Zhang, J. (2018). Health spending, savings and fertility in a lifecycle-dynastic model with longevity externalities. *Journal of Macroeconomics*.
- Zhao, K. (2014). Social security and the rise in health spending. *Journal of Monetary Economics*, 64:21–37.

Appendix A Model Appendix: Derivations and Optimality Conditions

This appendix collects two derivations behind the model of Section 3: the comparison of multiplicative and additive health amenities that motivates the specification of preferences, and the optimality conditions of the household problem.

A.1 The Amenity: Multiplicative versus Additive

Section 3.2 states that the multiplicative amenity creates an income gradient in the willingness to pay for health that an additive amenity lacks by construction. This appendix records the derivations behind that claim.

Scaling the GHH aggregator by $\Omega(h)$ makes the marginal utility of consumption depend on health,

$$u_{ch} = \Omega'(h) (c - v(\ell))^{-\gamma} \geq 0, \quad (52)$$

with strict inequality whenever $\Xi > 0$. The sign holds at every state, since $\Omega' \geq 0$ and $(c - v(\ell))^{-\gamma} > 0$ irrespective of the bracket in (5). Health is a state rather than a within-period choice, so this complementarity reaches the medical-spending decision indirectly: through u_h in the health envelope condition (58), which supplies the marginal continuation value in (56), and through the dependence of next period's u_c on h' in the consumption Euler equation (61).

Had the amenity entered the composite additively, as $u = [(c + s(h) - v(\ell))^{1-\gamma} - 1]/(1 - \gamma)$ with $s' > 0$, the same cross-partial would carry the opposite sign,

$$u_{ch} = -\gamma s'(h) (c + s(h) - v(\ell))^{-\gamma-1} < 0.$$

The sign of u_{ch} is not, however, the object that governs behavior. Medical spending responds to the marginal rate of substitution between health and consumption, which is what enters the medical-spending condition (56). Under the additive form that ratio is

$$\frac{u_h}{u_c} = \frac{s'(h) (c + s(h) - v(\ell))^{-\gamma}}{(c + s(h) - v(\ell))^{-\gamma}} = s'(h),$$

independent of consumption. Health and consumption are perfect substitutes inside the composite, so the willingness to pay for health carries no income gradient; the negative u_{ch} above is the concavity of the outer aggregator and by itself moves nothing. Under the multiplicative form the same ratio is

$$\frac{u_h}{u_c} = \frac{\Omega'(h)}{\Omega(h)} \left[\frac{z^{1-\gamma} - 1}{1 - \gamma} + \chi \right] z^\gamma, \quad z \equiv c - v(\ell),$$

which is increasing in c whenever $1 + \gamma [(z^{1-\gamma} - 1)/(1 - \gamma) + \chi] z^{\gamma-1} > 0$. The condition holds at every state at which the bracket in (5) is nonnegative, and can fail only inside the region where that bracket is negative, that is, at states the household ranks below not being alive.

The two forms also differ at the boundary of the feasible set. Because the amenity multiplies the aggregator, $u_c = \Omega(h) (c - v(\ell))^{-\gamma} \rightarrow \infty$ as $c \downarrow v(\ell)$, so consumption satisfies an Inada condition. Under an additive amenity the singularity sits instead at $c = v(\ell) - s(h)$, which is strictly negative whenever $s(h) > v(\ell)$. The Inada point is then unreachable, u_c stays finite as $c \rightarrow 0$, and a household can be close to indifferent between consuming and not.

A.2 Household Optimality Conditions

This appendix states the first-order, envelope, and Kuhn–Tucker conditions of the worker’s problem (32) and the retiree’s problem (37). Let λ_j denote the multiplier on the budget constraint, and recall that the two nodes h'_k of the health lottery (16) are functions of (m, h) through $\tilde{h}$. At an optimum at which the min and max operators in (16) are slack at both nodes, $\partial h'_k / \partial m = \partial H_j / \partial m$ and $\partial h'_k / \partial h = \partial \tilde{h} / \partial h$ for both k ,

with the derivatives given in (15). The interior first-order conditions are

$$\text{Consumption: } u_c(c, \ell, h) = \lambda_j(1 + \tau^c), \quad (53)$$

$$\text{Labor: } -u_\ell(c, \ell, h) = \lambda_j w v_j(h, \eta; g, \theta) (1 - \tau^\omega - \tau^p), \quad (54)$$

$$\text{Savings: } \lambda_j = \beta \sum_{k \in \{0,1\}} \Pr(k | j) \psi_{j+1}^{g,\theta}(h'_k) \mathbb{E}[V_{a,j+1}(a', h'_k, \eta'; g, \theta) | \eta], \quad (55)$$

$$\begin{aligned} \text{Medical spending: } \lambda_j(1 + \tau^m) = & \beta \frac{\partial H_j(m, h)}{\partial m} \sum_{k \in \{0,1\}} \Pr(k | j) \left\{ (\psi_{j+1}^{g,\theta})'(h'_k) \mathbb{E}[V_{j+1}(a', h'_k, \eta'; g, \theta) | \eta] \right. \\ & \left. + \psi_{j+1}^{g,\theta}(h'_k) \mathbb{E}[V_{h,j+1}(a', h'_k, \eta'; g, \theta) | \eta] \right\}. \end{aligned} \quad (56)$$

Condition (56) states the two returns to medical spending that the model keeps distinct—a survival return, through $(\psi_{j+1}^{g,\theta})'$ multiplying the level of the continuation value, and a health-capital return, through $\psi_{j+1}^{g,\theta}$ multiplying its marginal value in health—each averaged over the two nodes of the health lottery. Because ς shifts the struck node down by a constant, the marginal product of spending is common to the two nodes and factors out; what differs across nodes is the point at which the survival schedule and the continuation value are evaluated. If the floor $\underline{h}$ binds at the struck node, that node's health no longer responds to m and the $k = 1$ term drops from the right-hand side of (56).

Envelope conditions. For the worker's problem ($j < j_R$),

$$V_{a,j}(a, h, \eta; g, \theta) = \lambda_j \cdot (1 + r(1 - \tau^k)), \quad (57)$$

$$\begin{aligned} V_{h,j}(a, h, \eta; g, \theta) = & u_h(c, \ell, h) + \lambda_j w v_j(h, \eta; g, \theta) \varrho \ell (1 - \tau^\omega - \tau^p) \\ & + \beta \frac{\partial \tilde{h}}{\partial h} \sum_{k \in \{0,1\}} \Pr(k | j) \left\{ (\psi_{j+1}^{g,\theta})'(h'_k) \mathbb{E}[V_{j+1}(a', h'_k, \eta'; g, \theta) | \eta] \right. \\ & \left. + \psi_{j+1}^{g,\theta}(h'_k) \mathbb{E}[V_{h,j+1}(a', h'_k, \eta'; g, \theta) | \eta] \right\}, \end{aligned} \quad (58)$$

with $\partial \tilde{h} / \partial h$ from (15). The second term is the wage return to health. Its slope ϱ is common to the two skill groups, so any cross-group difference in the earnings value of health operates through the wage bill $w v_j \ell$ —through the intercept θ and the profile $e_j^{g,\theta}$ —rather than through a differential return: health is worth more, in earnings, to workers whose time is worth more. For the retiree's problem ($j \geq j_R$), $\ell = 0$ and that

term vanishes:

$$V_{a,j}^R(a, h, \eta; g, \theta) = \lambda_j \cdot (1 + r(1 - \tau^k)), \quad (59)$$

$$\begin{aligned} V_{h,j}^R(a, h, \eta; g, \theta) = & u_h(c, 0, h) \\ & + \beta \frac{\partial \tilde{h}}{\partial h} \sum_{k \in \{0,1\}} \Pr(k | j) \left\{ (\psi_{j+1}^{g,\theta})'(h'_k) \mathbb{E}[V_{j+1}(a', h'_k, \eta'; g, \theta) | \eta] \right. \\ & \left. + \psi_{j+1}^{g,\theta}(h'_k) \mathbb{E}[V_{h,j+1}(a', h'_k, \eta'; g, \theta) | \eta] \right\}. \quad (60) \end{aligned}$$

At the terminal age $j = J$, $\psi_{j+1}^{g,\theta} \equiv 0$ kills the continuation terms; the envelopes remain well defined since $a'_J = m_J = 0$.

Euler equation. Combining (55) with (57) one period ahead and (53),

$$u_c(c, \ell, h) = \beta (1 + r(1 - \tau^k)) \sum_{k \in \{0,1\}} \Pr(k | j) \psi_{j+1}^{g,\theta}(h'_k) \mathbb{E}[u_c(c'_k, \ell'_k, h'_k) | \eta], \quad (61)$$

where (c'_k, ℓ'_k) are next-period policies evaluated at the node- k health state.

Consumption–leisure condition. Combining (53) and (54),

$$\frac{-u_\ell(c, \ell, h)}{u_c(c, \ell, h)} = \frac{w v_j(h, \eta; g, \theta)(1 - \tau^\omega - \tau^p)}{1 + \tau^c}, \quad (62)$$

where the amenity cancels from the left-hand side, since $\Omega(h)$ scales u uniformly: $-u_\ell/u_c = v'(\ell)$.

Kuhn–Tucker for the borrowing constraint. The constraint $a' \geq 0$ generates a non-negative multiplier and turns the savings condition (55) into the weak inequality

$$\lambda_j \geq \beta \sum_{k \in \{0,1\}} \Pr(k | j) \psi_{j+1}^{g,\theta}(h'_k) \mathbb{E}[V_{a,j+1}(a', h'_k, \eta'; g, \theta) | \eta], \quad a' \geq 0, \quad (63)$$

with complementary slackness: the product of the gap and a' is zero.

Kuhn–Tucker for medical spending. The constraint $m \geq 0$ turns (56) into the weak inequality

$$\lambda_j(1 + \tau^m) \geq \beta \frac{\partial H_j(m, h)}{\partial m} \sum_{k \in \{0,1\}} \Pr(k | j) \left\{ (\psi_{j+1}^{g,\theta})'(h'_k) \mathbb{E}[V_{j+1}(\cdot) | \eta] + \psi_{j+1}^{g,\theta}(h'_k) \mathbb{E}[V_{h,j+1}(\cdot) | \eta] \right\}, \quad (64)$$

for $m \geq 0$, with complementary slackness. The cap $h'_k \leq \bar{h}$ in (16) is slack wherever $\bar{H}_j m^{\zeta_h} < \bar{h}$, by the asymptote property of (14); where it binds it introduces a non-

negative multiplier $\mu_{\bar{h}} \geq 0$ with complementary slackness

$$\mu_{\bar{h}} (\bar{h} - h'_0) = 0, \quad h'_0 \leq \bar{h}, \quad \mu_{\bar{h}} \geq 0, \quad (65)$$

entering the medical-spending condition as a subtracted term $\mu_{\bar{h}} \partial H_j / \partial m$ on the right-hand side of (64). At an interior optimum with $m > 0$ and both nodes strictly inside $(\underline{h}, \bar{h})$, all inequalities hold with equality and (56) is recovered.

Appendix B Estimation Details

This appendix supports the estimator of Section 4.4. It states the objective and the weighting choice, describes the algorithm that computes the equilibrium and the state space every reported number is computed on, and gives the reasons no standard errors are reported.

The estimator is deterministic. The state space is discretized and the stationary distribution over $(j, g, \theta, a, h, \eta)$ is computed by forward iteration, so every model moment is an exact distribution-weighted functional rather than a Monte Carlo average over simulated panels. The objective is therefore deterministic—re-evaluating it at the same parameter vector returns the same value—the asymptotic theory is that of classical minimum distance with no simulation-inflation term, and any non-smoothness the optimizer meets is a property of the model rather than of sampling noise. What remains is discretization: moments are kinked wherever mass crosses a grid cell.

Computing the equilibrium. The equilibrium of Definition 1 is the fixed point of three nested maps. The inner map is the household problem, solved by backward induction on the finite-horizon Bellman equations (32) and (37) together with the savings and medical-spending conditions (55)–(56) of Appendix A.2, with labor supply concentrated out through the intratemporal condition (62) and constrained maximization over the remaining controls (a', m) . The middle map iterates the population measure (47) jointly with the cohort weights (2) and the health-dependent survival schedule (3) to their mutual fixed point, which is condition 10 of the definition. The outer map adjusts the capital–labor ratio until the factor-price conditions (9)–(10) and the capital-market condition (26) clear, with the contribution rate and the stationary debt level moving through (23) and (20). That outer map is not nested inside the estimation objective: parameters are obtained at fixed anchor prices, as Section 4.4 describes, and the three maps run jointly only at the estimates and in the experiments. The endogenous grid method is not available here. Medical spending enters the continuation value non-separably and through its level, since the survival derivative in (56) multiplies $\mathbb{E}[V_{j+1}]$ rather than only its marginal value, and next-period health is itself a control. The end-of-period state is therefore not exogenous and the Euler equation cannot be inverted in

closed form. At the calibrated baseline the supporting residuals are small: the largest Euler-equation residual across interior cells is $10^{-7.5}$ in marginal-utility units, the goods and capital markets clear to within 8×10^{-5} of output and capital respectively, and the cohort-share invariant implied by (3) holds to 3×10^{-15} .

Objective and weighting. The estimator solves

$$\hat{\Theta}_{\text{MD}} = \arg \min_{\Theta} \sum_{k=1}^{10} \left(\frac{m_k^{\text{mod}}(\Theta) - m_k^{\text{dat}}}{m_k^{\text{dat}}} \right)^2 + \mathcal{P}(\Theta), \quad (66)$$

that is, minimum distance with $W = \text{diag}(1/(m^{\text{dat}})^2)$. The term $\mathcal{P}$ is a smooth penalty, zero over almost all of the admissible set, that excludes the region in which some household would assign negative value to survival; it binds nowhere near the estimates, where no household does. We do not use inverse-variance weighting, $W = \text{diag}(1/\text{SE}^2)$, for two reasons. The standard errors are not commensurable across this target set: one target is a convention with no sampling error at all, and the survey moments mix cluster-robust, delta-method, and bootstrap objects. And inverse-variance weighting turns the estimator into a matcher of t -statistics rather than of economic magnitudes—at the starting values it concentrated the objective on a single precisely estimated survey moment, leaving the moments that carry the paper’s mechanism nearly weightless. Equal proportional weighting prices a ten-per-cent miss identically wherever it occurs and carries no efficiency claim. Nine parameters against ten moments over-identifies the system by one, but the over-identifying restriction is the survival-anchor overlap described in Section 4.4 rather than a restriction the theory imposes on the data, so we report no over-identification test. Nor do we claim a numerical rank condition at the estimates: two parameters sit at boundaries of the admissible set, where only one-sided variation exists. Proportional weights on two survival levels near one make the sign of their small difference nearly costless.

Inference. We report point estimates without standard errors. The numerical derivatives that the minimum-distance sandwich formula requires are not stable here—the derivative of one of the two moments that dominate the objective at the estimates is not sign-stable across candidate step sizes, so a standard error built on it would be spuriously precise—and two estimates sit at boundaries of the admissible set, where the unconstrained sandwich does not apply in any case. The nine parameters are searched under a logit transform onto their admissible intervals, so a bound is an asymptote the search approaches without attaining. The two driven to the bottom of their intervals are the amenity scale, admissible on $[10^{-4}, 5]$, and the shock arrival rate, on $[0.01, 0.436]$. A step-size study and a treatment of the two boundary estimates are prerequisites to reporting inference rather than robustness checks on it. Until then the fit of Table 5, rather than a t -statistic, is the measure of precision on offer.

Optimizer and state space. The outer search is a derivative-free simplex method terminated by a no-further-improvement rule rather than by a formal convergence criterion. All reported numbers—estimates, fit, validation, and the experiments of Section 6—are computed on the coarser of two state-space grids. That grid carries 100 asset nodes on a geometric mesh reaching 300, ten health nodes, five nodes for the persistent productivity shock, and twenty for medical spending; the finer grid raises the last three to fifteen, seven and eighty. The productivity process (13) is discretized by the Rouwenhorst method, which matches the persistence of the underlying AR(1) exactly at this node count. The two grids differ in the levels of some moments, so reported levels carry the caveat while cross-economy comparisons, which the paper’s results rest on, do not. A polish on the finer grid awaits the health-block specification decision of Section 7.

Appendix C 2050 Demographic Inputs: Derivation and Sources

This appendix documents the single input that distinguishes the aged economy of Section 6 from the baseline: the 2050 old-age dependency ratio, its source, and the population growth rate that reproduces it.

Source. The demographic target for the 2050 experiments of Section 6 is the old-age dependency ratio $65+/(20-64)$ from CONAPO’s official population projections for México (Proyecciones de la Población de México y de las Entidades Federativas, 2023 revision; CONAPO, 2023): 0.3065 in 2050, against the 2020 census value of 0.1414 that disciplines the baseline (Section 4.3). The bracketing projections (0.1746 in 2030, 0.5064 in 2070) locate 2050 on a still-steepening path. The 2050 ratio comes from the same CONAPO release as the life tables used in the first step, The UN World Population Prospects is the natural robustness alternative and changes only the target number.

Method. The experiment imposes the 2050 population *structure*, not a demographic forecast path, and it does so through the single parameter the baseline already uses for the same purpose. In the stationary population, cohort weights satisfy (2), $\phi_{j+1}/\phi_j = \bar{\psi}_{j+1}/(1 + n_p)$, where n_p is the five-year growth rate (the annual rates quoted here are $(1 + n_p)^{1/5} - 1$) and $\bar{\psi}_{j+1}$ is the distribution-weighted average survival from band j to band $j + 1$ —an equilibrium object, since survival is health-modulated. We hold that schedule fixed at its baseline values and bisect n_p against the CONAPO target, which gives $n_{2050}^* = 1.483\%$ per year. Because the average survivals are themselves equilibrium objects, the bisection only proposes a growth rate; the confirming general-equilibrium solve is what we report. It realizes a dependency ratio of 0.30643, within

0.024% of the CONAPO value, and the pay-as-you-go identity evaluated at that ratio reproduces the reported contribution rate exactly.

Two arms not run. Section 6 explains why survival schedules and the birth composition stay at their 2020 values rather than moving to 2050 projections. Two further points complete that argument. A survival arm would need the CONAPO projection stripped to its exogenous component, since the model's own survival is health-modulated and a projected schedule already embeds behaviour. And an education-composition arm would need an external projection of educational attainment, without which it would conflate aging with skill upgrading.